

**Inverse Functions of q -Pochhammer Symbols,
Gaussian Coefficients, and Related q -Analogues**
Branch Geometry, Local Reversion, Singular Limits, and Computation

Research report developed from the ProveIt exponent- q -series corpus

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Abstract

The standard theory of basic hypergeometric functions is overwhelmingly a *forward* theory: products, recurrences, transformations, summations, limits, and coefficient identities are evaluated from their parameters. This report develops the complementary *inverse* theory. Given a level Y , we solve, branch by branch, equations such as

$$(a; q)_n = Y, \quad (a; q)_\infty = Y, \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_q = Y, \quad \Gamma_q(x) = Y,$$

not only for the displayed argument but also for the base q , a continuous order, and the remaining parameters. The distinction between a reciprocal and a compositional inverse is maintained throughout.

The main results include: an explicit branch atlas for finite and infinite q -Pochhammer symbols in the parameter a ; fifth-order reversion formulae at $a = 0$ and $q = 0$; all-order logarithmic inversion at $q = 1$; Puiseux laws at critical values, colliding factors, and roots of unity; Laurent–Puiseux inverses at $q = \pm\infty$; a radial root-of-unity expansion for $(a; q)_\infty$, including an explicit $q \rightarrow -1$ specialization; a log-periodically modulated inverse for large negative a ; inverse maps for complex order $(a; q)_\alpha$; positive-real global inverses of Gaussian polynomials; explicit expansions at $q = 0, 1, -1, \infty$; two real inverse branches of Γ_q at its unique minimum; saturating and two-sheeted inverse maps for the continuous q -binomial coefficient and q -beta function; and scaled inverse q -exponential expansions converging to the ordinary logarithm.

A branch-aware numerical method is supplied in a commented Python program. The experiments confirm the local formulae at 80-digit precision. The last chapters isolate conjectures on global critical-point geometry, root-of-unity monodromy, differential transcendence of generic inverse q -functions, and uniform double-scaling limits.

Scope and status

A statement labeled theorem, proposition, lemma, or corollary is proved in this report under the hypotheses printed with it. An “asymptotic inverse” is branchwise: a branch of every logarithm and fractional power must be fixed before inversion. Conjectures and research problems are explicitly marked. The numerical section is corroborative, not a substitute for proof.

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Chapter 1

Orientation, source map, and the inverse viewpoint

1.1 Relation to the existing exponent- q -series corpus

The source tree reviewed for this report contains two large LaTeX works. The first is a proof-oriented monograph on q -Pochhammer symbols, Gaussian coefficients, basic hypergeometric series, combinatorics, and arithmetic; the second is a consolidated frontier report connecting exponent sequences, q -series, infinite sinc products, and the Fabius–Rvachev system [15, 14]. We retain their notation and use as input their forward identities, especially

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a; q)_\infty = \prod_{j \geq 0} (1 - aq^j),$$

the finite q -binomial theorem, reciprocity of Gaussian coefficients, cyclotomic factorization, complex-order continuation, and the polylogarithmic expansion as $q \rightarrow 1$.

The present report does not reproduce those developments. Its organizing question is different:

For a fixed level Y , what are all local or global solutions in one chosen parameter, where do their branches meet, and what asymptotic coordinate makes the inverse regular?

In particular, “inverse q -Pochhammer” here never means merely $1/(a; q)_n$. It means a compositional or parameter inverse of a level map.

1.2 Four kinds of inverse problem

Let $F(z; \theta)$ be a q -analogue, where z is the active variable and θ denotes passive parameters. Four related but distinct problems occur.

Definition 1.1 (Branchwise compositional inverse). A local inverse branch at (z_0, Y_0) , with $Y_0 = F(z_0; \theta)$, is a germ $z = \mathcal{B}(Y; \theta)$ satisfying $F(\mathcal{B}(Y; \theta); \theta) = Y$ and $\mathcal{B}(Y_0; \theta) = z_0$.

Definition 1.2 (Parameter inverse). If a parameter θ_j is declared active, a parameter inverse solves $F(z; \theta_1, \dots, \theta_j, \dots) = Y$ for θ_j . The same analytic machinery applies, but the singular set changes because F_{θ_j} , rather than F_z , controls local invertibility.

A reciprocal $F^{-1} = 1/F$ is algebraically useful but is not a compositional inverse. Finally, a functional equation can possess a left-inverse or right-inverse operator without producing a pointwise inverse of a scalar parameter. Keeping these meanings separate prevents a common terminological ambiguity in the q -series literature.

1.3 Natural asymptotic coordinates

The active coordinate should be chosen before series reversion. The following table is used repeatedly.

Regime	regular coordinate	typical inverse scale
$q \rightarrow 0$	q	Taylor or generalized Puiseux
$q \rightarrow 1^-$	$t = -\log q$	Taylor in a logarithmic level
$q \rightarrow -1$ radially	$q = -e^{-t}$	Taylor in $1/\log Y$
$q \rightarrow \pm\infty$	$z = 1/q$	Laurent–Puiseux in $Y^{-1/d}$
$a \rightarrow \infty$, finite order	$1/a$	Laurent–Puiseux
$a \rightarrow -\infty$, infinite order	$L = \log(-a)$	square root of $\log Y$
$\alpha \rightarrow +\infty$	$r = q^\alpha$	logarithmic endpoint inverse
critical value	$(Y - Y_c)^{1/m}$	m -sheeted Puiseux

Chapter 2

Universal branchwise inversion calculus

2.1 Lagrange–Bürmann reversion

Theorem 2.1 (Analytic local inverse). *Let F be analytic near z_0 , put $Y_0 = F(z_0)$, and assume $F'(z_0) \neq 0$. Then a unique analytic inverse germ $z = \mathcal{B}(Y)$ exists near Y_0 . Its coefficients are*

$$\mathcal{B}(Y) = z_0 + \sum_{m \geq 1} \frac{(Y - Y_0)^m}{m!} \left[\frac{d^{m-1}}{dz^{m-1}} \left(\frac{z - z_0}{F(z) - Y_0} \right)^m \right]_{z=z_0}. \quad (2.1)$$

Proof. The analytic inverse theorem gives existence and uniqueness. For the coefficient of $(Y - Y_0)^m$, write a Cauchy integral around z_0 , change variables from $Y = F(z)$, and take the residue at z_0 . Integration by parts $m - 1$ times gives (2.1); this is the standard Lagrange–Bürmann formula [5, 7]. $\square$

For repeated use, suppose

$$w = c_1 z + c_2 z^2 + c_3 z^3 + c_4 z^4 + c_5 z^5 + O(z^6), \quad c_1 \neq 0.$$

Then direct reversion gives

$$\begin{aligned} z = & \frac{w}{c_1} - \frac{c_2}{c_1^3} w^2 + \frac{2c_2^2 - c_1 c_3}{c_1^5} w^3 \\ & + \frac{-c_1^2 c_4 + 5c_1 c_2 c_3 - 5c_2^3}{c_1^7} w^4 \\ & + \frac{-c_1^3 c_5 + 6c_1^2 c_2 c_4 + 3c_1^2 c_3^2 - 21c_1 c_2^2 c_3 + 14c_2^4}{c_1^9} w^5 + O(w^6). \end{aligned} \quad (2.2)$$

These coefficients are universal Bell-polynomial expressions. Formula (2.2) is the coefficient engine behind most explicit expansions below.

2.2 Implicit differentiation in passive parameters

Proposition 2.2 (Sensitivity of an inverse branch). *Let $F(z; \theta) = Y$, with $F_z \neq 0$, define a local branch $z = \mathcal{B}(Y; \theta)$. Then*

$$\mathcal{B}_Y = \frac{1}{F_z}, \quad \mathcal{B}_{\theta_j} = -\frac{F_{\theta_j}}{F_z}. \quad (2.3)$$

Moreover,

$$\mathcal{B}_{YY} = -\frac{F_{zz}}{F_z^3}, \quad (2.4)$$

and all mixed higher derivatives are obtained recursively by differentiating $F(\mathcal{B}(Y; \theta); \theta) = Y$.

Proof. Differentiate the identity once with respect to Y or θ_j . Differentiating $F_z \mathcal{B}_Y = 1$ once more in Y gives (2.4). Repetition yields multivariate Faà di Bruno polynomials. $\square$

For products, it is usually more stable to set $G = \log F$ and solve $G = \log Y$. Newton's correction then becomes

$$\Delta z = -\frac{G(z) - \text{Log } Y}{G_z(z)},$$

which avoids underflow when F is exponentially small.

2.3 Critical values and Puiseux inverses

Theorem 2.3 (Local critical-point normal form). *Suppose*

$$F(z) = Y_c + c_m(z - z_c)^m + c_{m+1}(z - z_c)^{m+1} + \cdots, \quad c_m \neq 0,$$

with $m \geq 2$. Then the inverse relation has exactly m local Puiseux branches

$$z - z_c = \omega \left(\frac{Y - Y_c}{c_m} \right)^{1/m} + O((Y - Y_c)^{2/m}), \quad \omega^m = 1. \quad (2.5)$$

A circuit around Y_c cyclically permutes these branches.

Proof. Factor $F(z) - Y_c = (z - z_c)^m H(z)$ with $H(z_c) = c_m \neq 0$. Choose an analytic branch of $H(z)^{1/m}$. Then $u = (z - z_c)H(z)^{1/m}$ is a local analytic coordinate with nonzero derivative and $F - Y_c = u^m$. Invert the coordinate and choose one of the m roots of $Y - Y_c$. $\square$

At a simple critical point $F'(z_c) = 0$, $F''(z_c) \neq 0$, the first terms can be made explicit. Write

$$\delta = F(z) - F(z_c) = a_2 s^2 + a_3 s^3 + a_4 s^4 + a_5 s^5 + \cdots, \quad s = z - z_c.$$

Then the two branches are

$$\begin{aligned} s = \pm \frac{\delta^{1/2}}{\sqrt{a_2}} - \frac{a_3}{2a_2^2} \delta \pm \frac{5a_3^2 - 4a_2 a_4}{8a_2^{7/2}} \delta^{3/2} \\ - \frac{a_2^2 a_5 - 3a_2 a_3 a_4 + 2a_3^3}{2a_2^5} \delta^2 + O(\delta^{5/2}). \end{aligned} \quad (2.6)$$

2.4 The discriminant and the branch atlas

For a polynomial level map $F(z; \theta) - Y$, its discriminant in z vanishes exactly when a root is multiple. Analytically, the corresponding critical-value set is

$$\text{Disc}_F = \{(Y, \theta) : \exists z, F(z; \theta) = Y, F_z(z; \theta) = 0\}.$$

The complement is a covering space on which inverse branches continue analytically. A *branch atlas* records: regular basepoints, critical values, local ramification indices, real monotonicity intervals, and branch continuation paths. This global object is more informative than one Taylor series.

Principle 2.4 (Endpoint saturation). *If a monotone level map approaches a finite limit exponentially, $F(z) = L(1 - c\rho^z + O(\rho^{2z}))$ with $0 < \rho < 1$, then its inverse has a logarithmic singularity:*

$$z = \frac{\log(1 - F/L) - \log c}{\log \rho} + O(1 - F/L).$$

If instead $\log F(z) \sim Cz^2$, then the inverse scale is $z \sim \sqrt{\log Y/C}$. Both mechanisms occur repeatedly below.

Chapter 3

Finite q -Pochhammer inverse in the parameter a

Throughout this chapter

$$P_n(a; q) := (a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad n \geq 1.$$

Unless explicitly specialized, $q \neq 0$.

3.1 Low orders and exact radicals

For $n = 1$, the inverse is $a = 1 - Y$. For $n = 2$,

$$Y = (1 - a)(1 - aq) = 1 - (1 + q)a + qa^2,$$

so, for $q \neq 0$,

$$a = \frac{1 + q \pm \sqrt{(1 - q)^2 + 4qY}}{2q}. \quad (3.1)$$

The discriminant identifies the unique finite critical value $Y_c = -(1 - q)^2/(4q)$. Formula (3.1) is the simplest model of the general two-sheeted branching at a real critical point.

3.2 The principal branch at $a = 0$

The finite q -binomial theorem gives

$$(a; q)_n = \sum_{r=0}^n (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q a^r. \quad (3.2)$$

Set

$$A = [n]_q, \quad B = q \begin{bmatrix} n \\ 2 \end{bmatrix}_q, \quad C = q^3 \begin{bmatrix} n \\ 3 \end{bmatrix}_q, \quad D = q^6 \begin{bmatrix} n \\ 4 \end{bmatrix}_q, \quad E = q^{10} \begin{bmatrix} n \\ 5 \end{bmatrix}_q.$$

Theorem 3.1 (Fifth-order inverse at $a = 0$). *Let $\delta = 1 - Y$ and assume $A \neq 0$. The inverse branch satisfying $a(1) = 0$ is*

$$\begin{aligned} a = & \frac{\delta}{A} + \frac{B}{A^3} \delta^2 + \left(\frac{2B^2}{A^5} - \frac{C}{A^4} \right) \delta^3 \\ & + \left(\frac{5B^3}{A^7} - \frac{5BC}{A^6} + \frac{D}{A^5} \right) \delta^4 \\ & + \left(\frac{14B^4}{A^9} - \frac{21B^2C}{A^8} + \frac{6BD + 3C^2}{A^7} - \frac{E}{A^6} \right) \delta^5 + O(\delta^6). \end{aligned} \quad (3.3)$$

Proof. Equation (3.2) gives $\delta = Aa - Ba^2 + Ca^3 - Da^4 + Ea^5 + O(a^6)$. Substitute $c_1 = A, c_2 = -B, c_3 = C, c_4 = -D, c_5 = E$ in (2.2). $\square$

The failure condition $A = [n]_q = 0$ is itself informative. It occurs when $q \neq 1$ is an n -th root of unity. The regular Taylor inverse then changes to a Puiseux inverse, analyzed in Section 3.5.

3.3 Inverse branches at the simple zeros

For generic q , the zeros are

$$a_j = q^{-j}, \quad 0 \leq j < n.$$

Proposition 3.2 (Derivative at a zero). *If the numbers q^{-j} are distinct, then*

$$P'_n(q^{-j}; q) = (-1)^{j+1} q^{-j(j-1)/2} (q; q)_j (q; q)_{n-1-j}. \quad (3.4)$$

Consequently, the inverse branch through $(Y, a) = (0, q^{-j})$ begins

$$\begin{aligned} a = q^{-j} &+ \frac{(-1)^{j+1} q^{j(j-1)/2}}{(q; q)_j (q; q)_{n-1-j}} Y \\ &+ \frac{L_j}{P'_n(q^{-j}; q)^2} Y^2 + O(Y^3), \end{aligned} \quad (3.5)$$

where

$$L_j = \sum_{\substack{0 \leq r < n \\ r \neq j}} \frac{q^r}{1 - q^{r-j}}.$$

Proof. Differentiate the vanishing factor $1 - aq^j$ and evaluate all remaining factors at $a = q^{-j}$. For $r < j$, use $1 - q^{r-j} = -q^{r-j}(1 - q^{j-r})$; the product of these powers is $q^{-j(j+1)/2}$. This gives (3.4). Writing $P = p_1 h + p_2 h^2/2 + \dots$, $h = a - q^{-j}$, gives $h = Y/p_1 - p_2 Y^2/(2p_1^3) + \dots$. If the vanishing factor is removed, logarithmic differentiation shows $p_2/p_1 = -2L_j$. $\square$

3.4 Complete real branch geometry

Theorem 3.3 (Real branch atlas). *Assume $q > 0, q \neq 1$, and order the positive zeros increasingly as $r_0 < r_1 < \dots < r_{n-1}$. Then:*

- (i) P_n is strictly decreasing and positive on $(-\infty, r_0)$, mapping that interval bijectively onto $(0, \infty)$.
- (ii) Every gap (r_j, r_{j+1}) contains exactly one critical point c_j , and it is simple: $P''_n(c_j) \neq 0$.
- (iii) There is no critical point in (r_{n-1}, ∞) .
- (iv) Every finite real branch collision occurs at one of the values $P_n(c_j)$ and has square-root form.

Proof. Away from the roots,

$$\frac{P'_n(a)}{P_n(a)} = \sum_{j=0}^{n-1} \frac{1}{a - r_j}, \quad \frac{d}{da} \frac{P'_n}{P_n} = - \sum_{j=0}^{n-1} \frac{1}{(a - r_j)^2} < 0. \quad (3.6)$$

On $(-\infty, r_0)$, every summand in the first expression is negative, so $P'_n < 0$; the endpoint limits are $+\infty$ and 0. In a gap, the logarithmic derivative decreases continuously from $+\infty$ to $-\infty$, giving one zero. At that zero, $P''_n/P_n = (P'_n/P_n)' < 0$, so the critical point is simple. To the right of the last root all summands are positive. The final assertion follows from Theorem 2.3. $\square$

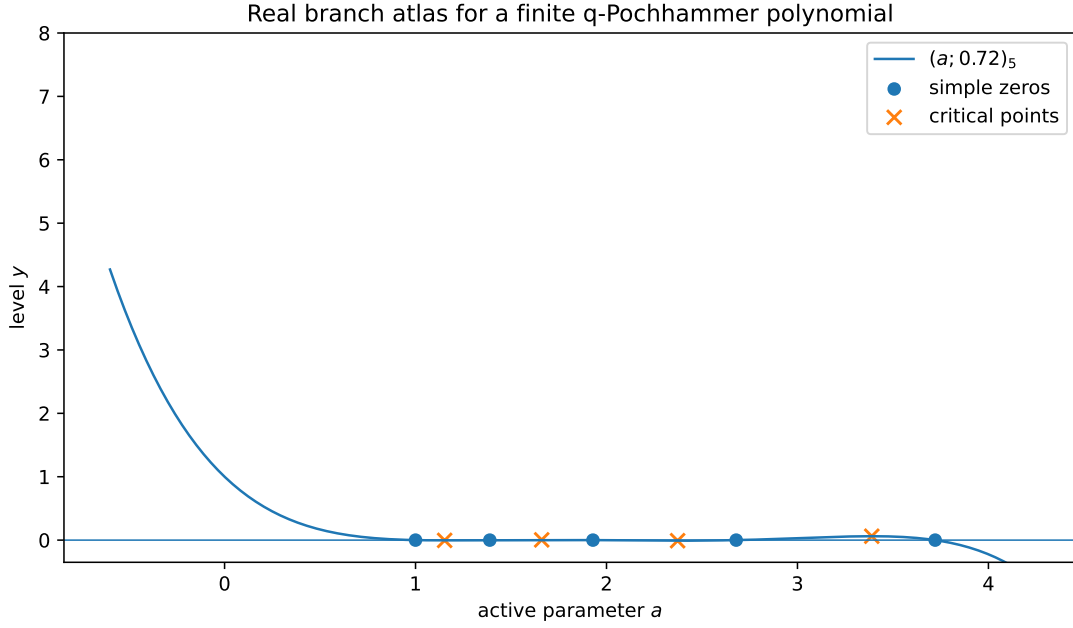


Figure 3.1: A finite q -Pochhammer polynomial has one simple critical point in every gap between consecutive positive zeros. Inverse branches meet in square-root pairs at the corresponding critical values.

3.5 Roots of unity and exact ramification

Theorem 3.4 (Root-of-unity factorization). *Let ζ be a primitive d -th root of unity and write $n = md + r$, $0 \leq r < d$. Then*

$$(a; \zeta)_n = (1 - a^d)^m (a; \zeta)_r. \quad (3.7)$$

If $r = 0$, the inverse at $a = 0, Y = 1$ is exactly

$$a = \omega(1 - Y^{1/m})^{1/d}, \quad \omega^d = 1. \quad (3.8)$$

If $r > 0$, the branch at $a = 0$ is regular because

$$\partial_a(a; \zeta)_n|_{a=0} = -\sum_{j=0}^{r-1} \zeta^j = -\frac{1 - \zeta^r}{1 - \zeta} \neq 0.$$

Proof. Group the product into m complete residue blocks modulo d , using $\prod_{j=0}^{d-1} (1 - a\zeta^j) = 1 - a^d$, followed by the residual block. When $r = 0$, solve $(1 - a^d)^m = Y$ directly. The derivative formula for $r > 0$ follows by differentiating at the origin. $\square$

At $q = -1$, this specializes to

$$(a; -1)_{2m} = (1 - a^2)^m, \quad (a; -1)_{2m+1} = (1 - a)(1 - a^2)^m.$$

Thus even order produces an exact square-root ramification at $a = 0$, while odd order remains locally univalent there.

3.6 Inverse at $a = \infty$

Let $N = \binom{n}{2}$, $A_n = (-1)^n q^N$, and define

$$e_1 = \sum_{j=0}^{n-1} q^{-j}, \quad e_2 = \sum_{0 \leq r < s < n} q^{-r-s}.$$

Factoring every term gives

$$P_n(a; q) = A_n a^n \prod_{j=0}^{n-1} (1 - q^{-j} a^{-1}).$$

Proposition 3.5 (Laurent–Puisseux inverse at infinity). *Choose a branch $Z = (Y/A_n)^{1/n}$ such that $Z/a \rightarrow 1$. Then*

$$a = Z + \frac{e_1}{n} + \left(\frac{(n-1)e_1^2}{2n^2} - \frac{e_2}{n} \right) Z^{-1} + O(Z^{-2}). \quad (3.9)$$

There are n branches at infinity, one for each n -th root.

Proof. Raise $\prod_j (1 - q^{-j}/a)$ to the power $1/n$, expand to second order in $1/a$, and revert $Z = a + s_1 + s_2/a + \dots$. $\square$

3.7 Cross-expansions of $a(Y; q)$ in the base

Fix a level Y and a branch $a_0 = 1 - Y$ at $q = 0$. Direct implicit expansion yields, for $n \geq 5$,

$$\begin{aligned} a(Y; q) &= a_0 + a_0(a_0 - 1)q + 2a_0^2(a_0 - 1)q^2 \\ &\quad + a_0^2(a_0 - 1)(5a_0 - 2)q^3 \\ &\quad + a_0^2(a_0 - 1)(2a_0 - 1)(7a_0 - 2)q^4 + O(q^5). \end{aligned} \quad (3.10)$$

For smaller n , terms are retained only until a missing factor $(1 - aq^j)$ first contributes.

Near $q = 1$, put $q = e^{-t}$ and choose an n -th root $r = Y^{1/n}$, so $a_0 = 1 - r$. If $r \neq 0$, then

$$\begin{aligned} a(Y; e^{-t}) &= a_0 + \frac{n-1}{2} a_0 t \\ &\quad + \frac{a_0(n-1)\{2n-4-3a_0(n-1)\}}{24(1-a_0)} t^2 + O(t^3). \end{aligned} \quad (3.11)$$

This follows either from [Theorem 2.2](#) or from the logarithmic expansion in the next chapter.

Chapter 4

Finite q -Pochhammer inverse in the base q

Fix a and n . As a function of q , $P_n(a; q)$ is a polynomial of degree $N = \binom{n}{2}$ when $a(1-a) \neq 0$. The generic level equation therefore has N complex branches. The purpose of this chapter is to choose the correct coordinates at $0, 1, \pm\infty$, roots, and critical points.

4.1 Derivatives, regular points, and the discriminant

For a nonzero product,

$$\partial_q \log P_n(a; q) = - \sum_{j=1}^{n-1} \frac{ajq^{j-1}}{1-aq^j}, \quad (4.1)$$

$$\partial_q^2 \log P_n(a; q) = - \sum_{j=1}^{n-1} \left(\frac{aj(j-1)q^{j-2}}{1-aq^j} + \frac{a^2 j^2 q^{2j-2}}{(1-aq^j)^2} \right). \quad (4.2)$$

At any base point q_0 where $P_q(a; q_0) \neq 0$, [Theorem 2.1](#) gives an ordinary Taylor inverse. The branch points are the critical values

$$Y_c = P_n(a; q_c), \quad P_q(a; q_c) = 0,$$

and the multiplicity is the order of the first nonzero q -derivative. Unlike the a -geometry in [Theorem 3.3](#), the real critical-point pattern in q need not be interlacing or uniform because the exponents j enter asymmetrically.

4.2 Expansion and inversion at $q = 0$

The factor independent of q may be removed:

$$\frac{P_n(a; q)}{1-a} = (aq; q)_{n-1} = \prod_{j=1}^{n-1} (1-aq^j). \quad (4.3)$$

Define

$$\eta = 1 - \frac{Y}{1-a}.$$

The coefficients of $\eta(q)$ count subsets of $\{1, \dots, n-1\}$ by their sum, weighted by powers of a . In particular, for $n \geq 6$,

$$\eta = aq + aq^2 + (a-a^2)q^3 + (a-a^2)q^4 + (a-2a^2)q^5 + O(q^6). \quad (4.4)$$

Theorem 4.1 (Fifth-order inverse at $q = 0$). *Assume $a \neq 0, 1$ and $n \geq 6$. Then the branch satisfying $q = 0$ at $Y = 1 - a$ is*

$$q = \frac{\eta}{a} - \frac{\eta^2}{a^2} + \frac{a+1}{a^3}\eta^3 - \frac{4a+1}{a^4}\eta^4 + \frac{3a^2+11a+1}{a^5}\eta^5 + O(\eta^6). \quad (4.5)$$

The same formula holds for $(a; q)_\infty$.

Proof. Expand (4.3). Through degree five, all contributing indices are at most five, so the coefficients stabilize once $n - 1 \geq 5$. Revert (4.4) with (2.2). The infinite product has the same finite jet because factors with $j \geq 6$ begin beyond the displayed order. $\square$

For arbitrary n , the exact coefficient engine is

$$[q^m] \frac{P_n(a; q)}{1-a} = \sum_{r \geq 0} (-a)^r \# \{S \subseteq \{1, \dots, n-1\} : |S| = r, \sum_{s \in S} s = m\}. \quad (4.6)$$

Inserting these coefficients into (2.2) produces the short-order corrections automatically.

4.3 All-order logarithmic expansion and inverse at $q = 1$

Put $q = e^{-t}$ and choose a branch of the logarithm. For fixed a with $a \notin [1, \infty)$,

$$\log(1 - ae^{-x}) = \log(1 - a) + \sum_{r \geq 1} \frac{(-1)^{r+1}}{r!} \text{Li}_{1-r}(a) x^r. \quad (4.7)$$

Let

$$S_r(n) = \sum_{j=0}^{n-1} j^r, \quad K_r(a, n) = (-1)^{r+1} \text{Li}_{1-r}(a) S_r(n).$$

Theorem 4.2 (Logarithmic $q = 1$ inverse). *Let*

$$u = \log \frac{Y}{(1-a)^n}.$$

If $K_1 \neq 0$, then the branch $q \rightarrow 1$ is obtained from

$$u = \sum_{r \geq 1} \frac{K_r}{r!} t^r, \quad q = e^{-t}, \quad (4.8)$$

and begins

$$t = \frac{u}{K_1} - \frac{K_2}{2K_1^3} u^2 + \frac{3K_2^2 - K_1 K_3}{6K_1^5} u^3 + \left(-\frac{K_4}{24K_1^5} + \frac{5K_2 K_3}{12K_1^6} - \frac{5K_2^3}{8K_1^7} \right) u^4 + O(u^5). \quad (4.9)$$

For $n \geq 2$ and $a \neq 0, 1$,

$$K_1 = \frac{a}{1-a} \binom{n}{2} \neq 0.$$

Proof. Substitute $x = jt$ in (4.7) and sum over $0 \leq j < n$, proving (4.8). Reversion gives (4.9). The expression for K_1 uses $\text{Li}_0(a) = a/(1-a)$ and $S_1(n) = \binom{n}{2}$. $\square$

The use of $t = -\log q$, rather than $q - 1$, is not cosmetic. It makes the coefficients polylogarithmic power sums, separates the base geometry from exponent combinatorics, and extends directly to the infinite product.

4.4 A critical scaled classical limit

The forward classical limit

$$G_n(x, t) = \frac{(e^{-xt}; e^{-t})_n}{(1 - e^{-t})^n} \longrightarrow (x)_n$$

contains a previously hidden inverse transition. The elementary expansion

$$\log \frac{1 - e^{-zt}}{zt} = -\frac{zt}{2} + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} z^{2m} t^{2m}$$

gives

$$\begin{aligned} \log \frac{G_n(x, t)}{(x)_n} = & -\frac{t}{2} \left(\sum_{j=0}^{n-1} (x+j) - n \right) \\ & + \sum_{m \geq 1} \frac{B_{2m} t^{2m}}{2m(2m)!} \left(\sum_{j=0}^{n-1} (x+j)^{2m} - n \right). \end{aligned} \quad (4.10)$$

Proposition 4.3 (Square-root inverse on the critical locus). *Assume $n \geq 2$, $(x)_n \neq 0$, and*

$$x = \frac{3-n}{2}.$$

Then the coefficient of t in (4.10) vanishes, while that of t^2 equals

$$\frac{n(n-1)(n+1)}{288} > 0.$$

Therefore inversion of $G_n(x, t) = Y$ at $t = 0$ has two square-root branches in $\log(Y/(x)_n)$.

Proof. The linear coefficient is zero exactly when $nx + n(n-1)/2 - n = 0$. On this locus,

$$\sum_{j=0}^{n-1} (x+j)^2 - n = \frac{n(n-1)(n+1)}{12},$$

and multiplication by $B_2/(2 \cdot 2!) = 1/24$ gives the stated number. Apply [Theorem 2.3](#). $\square$

For odd $n \geq 3$, the critical value of x is a nonpositive integer and one factor vanishes identically; the nontrivial normalized transition is therefore the even- n case. This is an example of a codimension-one parameter locus where a regular $q = 1$ inverse changes universality class.

4.5 Laurent–Puiseux inversion at $q = \pm\infty$

Assume $n \geq 2$, $a(1-a) \neq 0$, and set

$$N = \binom{n}{2}, \quad C = (1-a)(-a)^{n-1}.$$

Then

$$P_n(a; q) = Cq^N H(q^{-1}), \quad H(z) = \prod_{j=1}^{n-1} (1 - a^{-1}z^j). \quad (4.11)$$

Write $h_1 = -a^{-1}$, and

$$h_2 = \begin{cases} -a^{-1}, & n \geq 3, \\ 0, & n = 2. \end{cases}$$

Proposition 4.4 (Inverse at either infinite end). *Choose a branch*

$$Z = \left(\frac{Y}{C} \right)^{1/N}$$

for which $Z/q \rightarrow 1$. Then

$$q = Z + \frac{1}{Na} - \left(\frac{h_2}{N} + \frac{1}{2N} \left(\frac{1}{N} - 1 \right) h_1^2 \right) Z^{-1} + O(Z^{-2}). \quad (4.12)$$

The same formula describes $q \rightarrow +\infty$, $q \rightarrow -\infty$, and complex sectors; only the chosen N -th root changes.

Proof. From (4.11), $Z = qH(q^{-1})^{1/N} = q + s_1 + s_2/q + \dots$, where

$$s_1 = \frac{h_1}{N}, \quad s_2 = \frac{h_2}{N} + \frac{1}{2N} \left(\frac{1}{N} - 1 \right) h_1^2.$$

Reversion gives $q = Z - s_1 - s_2/Z + \dots$, and $-s_1 = 1/(Na)$. □

On the real line, parity determines the endpoints. If N is even, the same sign of Y/C is reached at both $+\infty$ and $-\infty$; if N is odd, the signs are opposite. Thus a large real level can have one or two real asymptotic branches depending on C , N , and the sign of Y .

4.6 Zeros in the base and colliding factors

Suppose $aq_0^j = 1$ for an index $1 \leq j < n$, and no other factor vanishes. Then

$$\partial_q P_n(a; q_0) = -jq_0^{-1} \prod_{\substack{0 \leq r < n \\ r \neq j}} (1 - q_0^{r-j}). \quad (4.13)$$

Therefore

$$q = q_0 + \frac{Y}{\partial_q P_n(a; q_0)} + O(Y^2).$$

If exactly m indices satisfy $aq_0^j = 1$, then the zero has multiplicity m and the inverse has m -th-root ramification. Such collisions occur precisely when ratios of relevant exponents are compatible with a root of unity. They are the base-variable analogue of [Theorem 3.4](#).

At a nonzero value $P_n(a; -1)$, the point $q = -1$ is ordinarily regular; its first inverse coefficient is obtained by substituting -1 in (4.1). Exceptional values of a for which this logarithmic derivative vanishes create square-root branches. The Gaussian coefficient has a much more rigid $q = -1$ theory, developed in [Chapter 7](#).

Chapter 5

The infinite q -Pochhammer inverse

For $0 < |q| < 1$, define

$$P_\infty(a; q) = (a; q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

The forward product is entire in a , while its behavior in q becomes singular on the unit circle. Both facts are reflected sharply in its inverse branches.

5.1 Inverse in a : origin, zeros, and real branches

The q -binomial theorem gives

$$(a; q)_\infty = \sum_{r \geq 0} \frac{(-1)^r q^{\binom{r}{2}}}{(q; q)_r} a^r.$$

Taking the limit $n \rightarrow \infty$ in [Theorem 3.1](#) gives a convergent local inverse at $Y = 1$. Its first four terms simplify to

$$\begin{aligned} a &= (1 - q)\delta + \frac{q(1 - q)}{1 + q}\delta^2 \\ &\quad + \frac{q^2(1 - q)(q^2 + q + 2)}{(1 + q)^2(1 + q + q^2)}\delta^3 \\ &\quad + \frac{q^3(1 - q)(q^4 + q^3 + 5q^2 + 5)}{(1 + q)^3(1 + q^2)(1 + q + q^2)}\delta^4 + O(\delta^5), \end{aligned} \tag{5.1}$$

where $\delta = 1 - Y$.

The zeros are again $a_j = q^{-j}$, now for all $j \geq 0$. They are simple when q is not a root of unity, and

$$\partial_a(a; q)_\infty \big|_{a=q^{-j}} = (-1)^{j+1} q^{-j(j-1)/2} (q; q)_j (q; q)_\infty. \tag{5.2}$$

Thus the zero-branch expansion in [Theorem 3.2](#) remains valid after replacing $(q; q)_{n-1-j}$ by $(q; q)_\infty$.

Theorem 5.1 (Real branch atlas for the entire product). *Let $0 < q < 1$. On $(-\infty, 1)$, $(a; q)_\infty$ is positive and strictly decreasing from $+\infty$ to 0, so it has a unique positive level inverse there. Between every pair $q^{-j} < q^{-(j+1)}$ there is exactly one simple critical point, and the associated inverse branches meet with square-root ramification.*

Proof. The normally convergent logarithmic derivative is

$$\partial_a \log(a; q)_\infty = \sum_{j \geq 0} \frac{1}{a - q^{-j}},$$

and its derivative is the strictly negative sum of squares. The proof of [Theorem 3.3](#) passes to the infinite product on every compact interval avoiding the zeros. As $a = -x \rightarrow -\infty$, infinitely many factors exceed one and the product diverges; a quantitative form follows next. $\square$

5.2 Large negative a and log-periodic inversion

Set $q = e^{-\tau}$, $\tau > 0$, and $a = -x$, $x > 0$. Write uniquely

$$x = q^{-m}r, \quad m \in \mathbb{N}_0, \quad 1 \leq r < q^{-1},$$

or equivalently $L = \log x = m\tau + \rho$, $0 \leq \rho < \tau$. Splitting the product at $j = m$ gives the exact identity

$$(-x; q)_\infty = r^m q^{-m(m+1)/2} (-q/r; q)_m (-r; q)_\infty. \quad (5.3)$$

Define

$$\Psi_q(\rho) = \log[(-e^\rho; q)_\infty (-qe^{-\rho}; q)_\infty] - \frac{\rho}{2} - \frac{\rho^2}{2\tau}. \quad (5.4)$$

Theorem 5.2 (Theta-modulated growth). *The function Ψ_q extends to a smooth τ -periodic function, and as $L \rightarrow +\infty$,*

$$\log(-e^L; q)_\infty = \frac{L^2}{2\tau} + \frac{L}{2} + \Psi_q(L \bmod \tau) + O(e^{-L}). \quad (5.5)$$

Consequently, if $Y = (-x; q)_\infty \rightarrow \infty$, put

$$R = \sqrt{2\tau \log Y}, \quad \rho_0 = (R - \tau/2) \bmod \tau.$$

Then the inverse satisfies

$$\log x = R - \frac{\tau}{2} + \frac{\tau(\tau/8 - \Psi_q(\rho_0))}{R} + O((\log Y)^{-1}). \quad (5.6)$$

Proof. In (5.3), replace the finite product $(-q/r; q)_m$ by $(-q/r; q)_\infty$; the omitted tail is $1 + O(q^m) = 1 + O(e^{-L})$. Since $m = (L - \rho)/\tau$, elementary algebra gives

$$m\rho + \frac{\tau m(m+1)}{2} = \frac{L^2}{2\tau} + \frac{L}{2} - \frac{\rho^2}{2\tau} - \frac{\rho}{2}.$$

This proves (5.5). Under $\rho \mapsto \rho + \tau$, the product inside the logarithm changes by the factor $e^{\rho+\tau}$; the two subtracted quadratic terms change by the same amount, proving periodicity. Complete the square in (5.5) and expand the positive square root. The phase correction is only $O(R^{-1})$, so replacing the exact phase by ρ_0 changes the displayed term by $O(R^{-2})$. $\square$

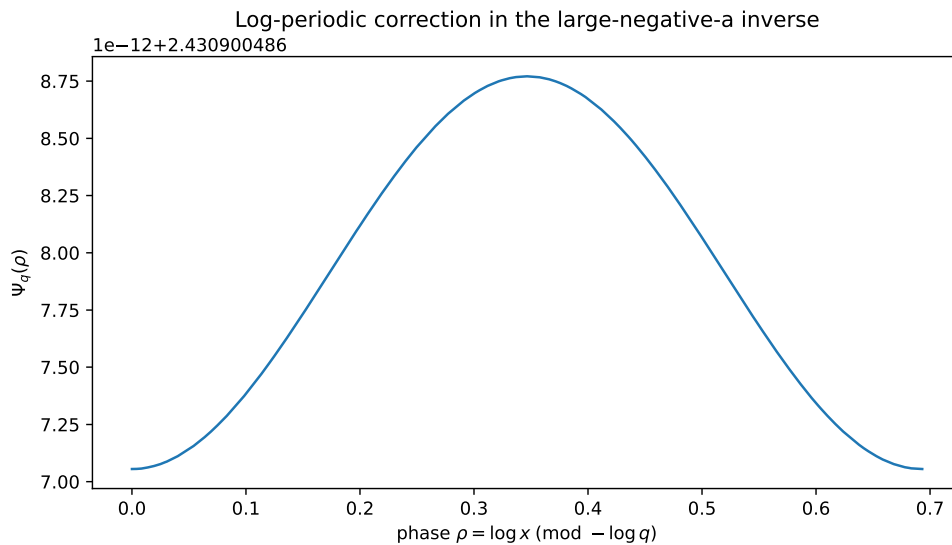


Figure 5.1: The bounded theta-type function Ψ_q produces a genuine log-periodic modulation in the inverse of $(-x; q)_\infty$. It is invisible in a purely smooth approximation in powers of $1/\log x$.

This phenomenon is structurally close to periodic fluctuations in digital and divide-and-conquer asymptotics. It also supplies a new bridge to the exponent-sequence themes in the Fabius–Rvachev corpus: a geometric scale induces periodicity after taking a logarithm.

5.3 Radial approach to a root of unity

The Lambert-series identity

$$\log(a; q)_\infty = - \sum_{m \geq 1} \frac{a^m}{m(1 - q^m)} \quad (5.7)$$

permits a uniform treatment of $q = 1, -1$, and every root of unity. Let ζ be a primitive d -th root and put $q = \zeta e^{-t}$, $t \rightarrow 0^+$. Assume $|a| < 1$, so all rearrangements are absolutely convergent.

Theorem 5.3 (Radial root-of-unity expansion). *Define*

$$A_d(a) = \frac{1}{d^2} \text{Li}_2(a^d), \quad (5.8)$$

$$C_{d,\zeta}(a) = \frac{1}{2d} \log(1 - a^d) - \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m}{m(1 - \zeta^m)}, \quad (5.9)$$

$$T_{d,\zeta}(a) = -\frac{a^d}{12(1 - a^d)} + \sum_{\substack{m \geq 1 \\ d \nmid m}} \frac{a^m \zeta^m}{(1 - \zeta^m)^2}. \quad (5.10)$$

Then

$$\log(a; \zeta e^{-t})_\infty = -\frac{A_d(a)}{t} + C_{d,\zeta}(a) + T_{d,\zeta}(a)t + O(t^2). \quad (5.11)$$

If $A_d(a) \neq 0$, set $X = -\text{Log } Y + C_{d,\zeta}(a)$ and $c_{d,\zeta} = -T_{d,\zeta}$. The radial inverse begins

$$t = \frac{A_d}{X} + \frac{A_d^2 c_{d,\zeta}}{X^3} + O(X^{-4}). \quad (5.12)$$

Proof. Split (5.7) into multiples $m = d\ell$ and nonmultiples. For the multiples, use

$$\frac{1}{1 - e^{-d\ell t}} = \frac{1}{d\ell t} + \frac{1}{2} + \frac{d\ell t}{12} + O(t^3).$$

The three resulting sums are respectively $-\text{Li}_2(a^d)/(d^2 t)$, $(2d)^{-1} \log(1 - a^d)$, and $-a^d t/[12(1 - a^d)]$. For $d \nmid m$, expand the denominator analytically at $t = 0$; its first derivative is $-m\zeta^m/(1 - \zeta^m)^2$, producing the second sums in (5.9)–(5.10). Reversion of $X = A_d/t + c_{d,\zeta}t + O(t^2)$ proves (5.12). $\square$

5.3.1 The specialization $q \rightarrow 1^-$

For $d = 1$, $\zeta = 1$, the theorem gives the familiar polylogarithmic expansion, here extended far enough to invert:

$$\begin{aligned} \log(a; e^{-t})_\infty = & -\frac{\text{Li}_2(a)}{t} + \frac{1}{2} \log(1 - a) - \frac{a}{12(1 - a)} t \\ & + \frac{a(1 + a)}{720(1 - a)^3} t^3 + O(t^5). \end{aligned} \quad (5.13)$$

Let

$$A = \text{Li}_2(a), \quad c = \frac{a}{12(1 - a)}, \quad d = \frac{a(1 + a)}{720(1 - a)^3}, \quad X = -\log Y + \frac{1}{2} \log(1 - a).$$

Then

$$t = \frac{A}{X} + \frac{A^2 c}{X^3} + \frac{2A^3 c^2 - A^4 d}{X^5} + O(X^{-7}), \quad q = e^{-t}. \quad (5.14)$$

For $0 < a < 1$, this is a real inverse as $Y \downarrow 0$.

5.3.2 The specialization $q \rightarrow -1$

For $d = 2$, $\zeta = -1$, the constants simplify:

$$\begin{aligned} \log(a; -e^{-t})_{\infty} = & -\frac{\text{Li}_2(a^2)}{4t} + \frac{1}{2} \log(1-a) - \frac{a(a+3)}{12(1-a^2)}t \\ & + \frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3}t^3 + O(t^5). \end{aligned} \quad (5.15)$$

The odd powers arise from separating even and odd Lambert indices. With

$$A_- = \frac{1}{4} \text{Li}_2(a^2), \quad c_- = \frac{a(a+3)}{12(1-a^2)}, \quad d_- = \frac{a(15a^4 + 4a^3 + 90a^2 + 4a + 15)}{720(1-a^2)^3},$$

and the same $X = -\log Y + \frac{1}{2} \log(1-a)$,

$$t = \frac{A_-}{X} + \frac{A_-^2 c_-}{X^3} + \frac{2A_-^3 c_-^2 - A_-^4 d_-}{X^5} + O(X^{-7}), \quad q = -e^{-t}. \quad (5.16)$$

This is the first nontrivial real root-of-unity inverse treated beyond $q = 1$ in the present atlas.

5.4 Expansion at $q = 0$

The local expansion in [Theorem 4.1](#) applies without change. It is often preferable to use the exact identity

$$\frac{(a; q)_{\infty}}{1-a} = (aq; q)_{\infty}$$

and then evaluate the inverse series in the bounded variable $\eta = 1 - Y/(1-a)$. This avoids cancellation when a is close to one.

5.5 Complex branch remarks

For a complex target, $X = -\text{Log } Y + C$ in (5.12) depends on the logarithm branch. Each addition of $2\pi i$ to $\text{Log } Y$ produces a different asymptotic sheet. The root-of-unity point is therefore an essential accumulation point of inverse branches, not an isolated algebraic branch point. Sectorial error bounds can be obtained by keeping t in a cone where $\Re t > 0$ and using the same Lambert-series splitting.

Chapter 6

Inverse with respect to continuous order and geometric step

6.1 Complex order

Fix a branch of $\text{Log } q$, with $0 < |q| < 1$, and define

$$(a; q)_\alpha = \frac{(a; q)_\infty}{(aq^\alpha; q)_\infty}, \quad q^\alpha = e^{\alpha \text{Log } q}. \quad (6.1)$$

This agrees with the finite product when $\alpha \in \mathbb{N}_0$. On a simply connected logarithmic domain satisfying $|a| < 1$ and $|aq^\alpha| < 1$, the convergent Lambert representation is

$$\log(a; q)_\alpha = - \sum_{m \geq 1} \frac{a^m (1 - q^{m\alpha})}{m(1 - q^m)}. \quad (6.2)$$

Beyond that domain, the identity and the inverse formulae below are understood by meromorphic continuation on any simply connected region avoiding zeros and poles and carrying compatible logarithm branches.

Theorem 6.1 (Local inverse in the order). *Fix α_0 away from the poles of $(a; q)_\alpha$, assume $Y_0 = (a; q)_{\alpha_0} \neq 0$ and $|aq^{\alpha_0}| < 1$ so that the Lambert cumulants below converge, put $s = \alpha - \alpha_0$, and*

$$u = \text{Log}(Y/Y_0).$$

Define

$$\lambda_r = (\text{Log } q)^r \sum_{m \geq 1} \frac{m^{r-1} a^m q^{m\alpha_0}}{1 - q^m}, \quad r \geq 1. \quad (6.3)$$

Then

$$u = \sum_{r \geq 1} \frac{\lambda_r}{r!} s^r. \quad (6.4)$$

If $\lambda_1 \neq 0$, the inverse branch is

$$\begin{aligned} s = & \frac{u}{\lambda_1} - \frac{\lambda_2}{2\lambda_1^3} u^2 + \frac{3\lambda_2^2 - \lambda_1 \lambda_3}{6\lambda_1^5} u^3 \\ & + \left(-\frac{\lambda_4}{24\lambda_1^5} + \frac{5\lambda_2 \lambda_3}{12\lambda_1^6} - \frac{5\lambda_2^3}{8\lambda_1^7} \right) u^4 + O(u^5). \end{aligned} \quad (6.5)$$

If the first nonzero λ_r has index $r = m > 1$, the order inverse has m -th-root Puiseux ramification.

Proof. Use the local quotient

$$\frac{(a; q)_{\alpha_0+s}}{(a; q)_{\alpha_0}} = \frac{(aq^{\alpha_0}; q)_{\infty}}{(aq^{\alpha_0+s}; q)_{\infty}}.$$

Its logarithm is a normally convergent Lambert series under the stated hypothesis. Expanding $q^{ms} = \exp(ms \log q)$ and interchanging the sums gives (6.4). Apply (2.2), or Theorem 2.3 in the degenerate case. $\square$

At $\alpha_0 = 0$, provided $|a| < 1$, $Y_0 = 1$ and

$$\lambda_r = (\log q)^r \sum_{m \geq 1} \frac{m^{r-1} a^m}{1 - q^m}.$$

Thus the inverse order is governed by a hierarchy of Lambert series. At a positive integer N , the same formula simply inserts q^{mN} ; the series then converges rapidly whenever $|aq^N| < 1$.

6.2 The endpoint $\alpha \rightarrow +\infty$

Assume $0 < q < 1$, $a \neq 0$, $(a; q)_{\infty} \neq 0$, and let $\alpha \rightarrow +\infty$. Put

$$b = aq^{\alpha}, \quad R = \frac{Y}{(a; q)_{\infty}}, \quad r = R - 1.$$

Then

$$R = \frac{1}{(b; q)_{\infty}} = e_q(b) = 1 + \frac{b}{1 - q} + \frac{b^2}{(1 - q)(1 - q^2)} + \cdots.$$

Proposition 6.2 (Logarithmic endpoint inverse). *As $R \rightarrow 1$,*

$$\begin{aligned} b &= (1 - q)r - \frac{1 - q}{1 + q}r^2 \\ &\quad + \frac{(1 - q)(2q^2 + q + 1)}{(1 + q)^2(1 + q + q^2)}r^3 + O(r^4), \end{aligned} \tag{6.6}$$

and therefore

$$\begin{aligned} \alpha &= \frac{1}{\log q} \left[\log \frac{(1 - q)r}{a} - \frac{r}{1 + q} \right. \\ &\quad \left. + \frac{3q^2 + q + 1}{2(1 + q)^2(1 + q + q^2)}r^2 + O(r^3) \right]. \end{aligned} \tag{6.7}$$

In particular, the inverse order diverges logarithmically as the level approaches $(a; q)_{\infty}$.

Proof. Revert the Taylor series of $e_q(b) - 1$, then use $\alpha = \log(b/a)/\log q$. Expanding the logarithm of (6.6) gives (6.7). $\square$

The endpoint is saturating: a finite change in α multiplies the small parameter b by q . This is precisely the mechanism in Theorem 2.4.

6.3 Poles in the order variable

The denominator in (6.1) vanishes when

$$aq^{\alpha_0+j} = 1, \quad j \in \mathbb{N}_0.$$

Suppose only one factor vanishes and the numerator does not. If

$$D_j = \frac{d}{d\alpha}(aq^\alpha; q)_\infty \Big|_{\alpha=\alpha_0},$$

then

$$(a; q)_\alpha = \frac{(a; q)_\infty}{D_j(\alpha - \alpha_0)} + O(1).$$

Hence the inverse at large level is an ordinary reciprocal expansion,

$$\alpha = \alpha_0 + \frac{(a; q)_\infty}{D_j Y} + O(Y^{-2}). \quad (6.8)$$

For a fixed base $0 < |q| < 1$, distinct denominator factors cannot collide. Under continuation of the base toward a root of unity, however, several factors may coalesce; the pole order then increases and the inverse scale becomes $Y^{-1/m}$. Thus the order variable exhibits the same root-of-unity collision mechanism as the base variable, but at poles rather than zeros.

6.4 Inversion in the geometric step

Consider the two-base symbol

$$F(\beta) = (a; q^\beta)_\alpha.$$

Fixing a branch of $\text{Log } q$, set $Q = q^\beta$. Every inverse result in the base Q immediately gives

$$\beta = \frac{\text{Log } Q}{\text{Log } q}. \quad (6.9)$$

This simple reduction is preferable to differentiating a nested exponential. For direct sensitivity calculations,

$$\frac{d}{d\beta} \log(a; q^\beta)_n = -(\text{Log } q) \sum_{j=1}^{n-1} \frac{ajq^{\beta j}}{1 - aq^{\beta j}},$$

which is the q -derivative formula (4.1) multiplied by the Jacobian $Q \text{Log } q$.

More generally, for a signed product

$$\mathcal{P}(q) = \prod_{\nu=1}^r (a_\nu; q)^{\sigma_\nu}_{\alpha_\nu}, \quad \sigma_\nu \in \mathbb{Z},$$

all logarithmic cumulants add. Thus the local inverse coefficients are obtained by replacing every K_r , λ_r , or Lambert coefficient by the corresponding signed sum. This observation covers individual terms of basic hypergeometric series, q -beta ratios, and many partition products without a new inversion proof.

Chapter 7

Gaussian coefficients: inverse in the base

Let $0 < k < n$ and

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{j=1}^k \frac{1 - q^{n-k+j}}{1 - q^j}.$$

It is a palindromic polynomial of degree

$$d = k(n - k),$$

with nonnegative integral coefficients. The coefficient of q^m counts partitions of m inside a $k \times (n - k)$ rectangle [2, 16].

7.1 The positive real inverse is global

Theorem 7.1 (Unique positive-real Gaussian inverse). *For $0 < k < n$, $G_{n,k}(q)$ is strictly increasing on $(0, \infty)$, with*

$$G_{n,k}(0) = 1, \quad G_{n,k}(q) \rightarrow \infty \quad (q \rightarrow \infty).$$

Consequently, every level $Y > 1$ has a unique positive real inverse $q = Q_{n,k}(Y)$.

Proof. The polynomial has nonnegative coefficients and its coefficient of q is one. Hence its derivative is positive for every $q > 0$. Its leading coefficient is one and its degree is $d > 0$. $\square$

This theorem supplies an unusually clean global branch. Negative and complex branches are organized by cyclotomic zeros and critical values.

7.2 The partition reversion at $q = 0$

Write

$$G_{n,k}(q) = 1 + q + c_2 q^2 + c_3 q^3 + c_4 q^4 + c_5 q^5 + \cdots,$$

where c_j is the number of partitions of j fitting in the rectangle.

Theorem 7.2 (Fifth-order Gaussian inverse at zero). *Let $\delta = Y - 1$. Then the positive branch has*

$$\begin{aligned} q = & \delta - c_2 \delta^2 + (2c_2^2 - c_3) \delta^3 \\ & + (-5c_2^3 + 5c_2 c_3 - c_4) \delta^4 \\ & + (14c_2^4 - 21c_2^2 c_3 + 6c_2 c_4 + 3c_3^2 - c_5) \delta^5 + O(\delta^6). \end{aligned} \tag{7.1}$$

If $\min(k, n - k) \geq 5$, then $(c_2, c_3, c_4, c_5) = (2, 3, 5, 7)$, and

$$q = \delta - 2\delta^2 + 5\delta^3 - 15\delta^4 + 52\delta^5 + O(\delta^6). \tag{7.2}$$

Proof. Apply (2.2) with $c_1 = 1$. If both rectangle dimensions are at least five, every partition of an integer at most five fits, so the coefficients are the ordinary partition numbers $p(2), \dots, p(5)$. $\square$

The coefficients in (7.2) are themselves a natural inverse-partition sequence. Determining their asymptotics when the rectangle is unbounded is a well-posed combinatorial problem discussed in Theorem 14.6.

7.3 Expansion at $q = 1$

Put $q = e^t$, and let

$$u = \log \frac{Y}{\binom{n}{k}}.$$

From

$$\log \frac{e^{zt} - 1}{zt} = \frac{zt}{2} + \sum_{r \geq 1} \frac{B_{2r}}{2r(2r)!} z^{2r} t^{2r},$$

one obtains

$$u = \frac{d}{2}t + \sum_{r \geq 1} \frac{B_{2r} t^{2r}}{2r(2r)!} \sum_{j=1}^k ((n-k+j)^{2r} - j^{2r}). \quad (7.3)$$

The absence of odd centered powers is the analytic form of palindromicity.

Theorem 7.3 (Gaussian inverse at one). As $Y \rightarrow \binom{n}{k}$,

$$t = \frac{2u}{d} - \frac{n+1}{3d^2} u^2 + \frac{(n+1)^2}{9d^3} u^3 + O(u^4), \quad (7.4)$$

$$q = 1 + \frac{2u}{d} + \frac{5-n}{3d^2} u^2 + \frac{n^2-4n+7}{9d^3} u^3 + O(u^4). \quad (7.5)$$

Proof. The quadratic coefficient in (7.3) is

$$\frac{1}{24} \sum_{j=1}^k ((n-k+j)^2 - j^2) = \frac{d(n+1)}{24}.$$

There is no cubic term. Revert $u = (d/2)t + d(n+1)t^2/24 + O(t^4)$, then expand $q = e^t$. $\square$

For $0 < q < 1$, the theorem's variable $t = \log q$ is negative. If one prefers $\tau = -\log q > 0$, substitute $t = -\tau$ in both the forward and inverse series. The formula itself is analytic across $q = 1$.

7.4 Reciprocity and the inverse at infinity

Palindromicity is the exact identity

$$G_{n,k}(q) = q^d G_{n,k}(q^{-1}). \quad (7.6)$$

Let $Z = Y^{1/d}$ be chosen so that $Z/q \rightarrow 1$. If $c_2 = [q^2]G_{n,k}$, then

$$q = Z - \frac{1}{d} - \left(\frac{c_2}{d} - \frac{d-1}{2d^2} \right) Z^{-1} + O(Z^{-2}). \quad (7.7)$$

This is the Gaussian specialization of Theorem 4.4. The same branch formula works at $q = -\infty$ after choosing $Z \sim q$. If d is even, a large positive level has both positive and negative asymptotic candidates; if d is odd, the negative branch corresponds to a large negative level.

7.5 The point $q = -1$

The q -Lucas theorem or cyclotomic factorization gives

$$G_{n,k}(-1) = \begin{cases} 0, & n = 2m, k = 2r + 1, \\ \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \text{otherwise.} \end{cases} \quad (7.8)$$

Theorem 7.4 (Inverse branches at $q = -1$). *Let $d = k(n - k)$.*

(i) *If $n = 2m, k = 2r + 1$, then $q = -1$ is a simple zero and*

$$G'_{2m,2r+1}(-1) = (m - r) \binom{m}{r}. \quad (7.9)$$

Hence

$$q = -1 + \frac{Y}{(m - r) \binom{m}{r}} + O(Y^2).$$

(ii) *Otherwise let $Y_0 = G_{n,k}(-1) \neq 0$. Then d is even and*

$$G'_{n,k}(-1) = -\frac{d}{2} Y_0. \quad (7.10)$$

Therefore

$$q = -1 - \frac{2(Y - Y_0)}{dY_0} + O((Y - Y_0)^2).$$

Proof. In case (i), the factor $\Phi_2(q) = q + 1$ occurs exactly once. Dividing the product formula by $q + 1$ and taking the limit, or applying q -Lucas to the quotient, gives (7.9). In case (ii), differentiate reciprocity (7.6):

$$G'(q) = dq^{d-1}G(q^{-1}) - q^{d-2}G'(q^{-1}).$$

At $q = -1$ and even d , this becomes $G' = -dG - G'$, proving (7.10). $\square$

7.6 General roots of unity

The cyclotomic factorization is

$$G_{n,k}(q) = \prod_{s \geq 1} \Phi_s(q)^{e_s}, \quad e_s = \left\lfloor \frac{n}{s} \right\rfloor - \left\lfloor \frac{k}{s} \right\rfloor - \left\lfloor \frac{n-k}{s} \right\rfloor \in \{0, 1\}. \quad (7.11)$$

The exponent e_s is a carry indicator: it equals one precisely when adding the residues of k and $n - k$ modulo s crosses s .

Corollary 7.5 (Cyclotomic inverse atlas). *Let ζ be a primitive s -th root of unity. If $e_s = 1$, then $G_{n,k}$ has a simple zero at ζ , and the inverse at $Y = 0$ is analytic with slope*

$$\frac{1}{G'_{n,k}(\zeta)} = \frac{1}{\Phi'_s(\zeta) \prod_{r \neq s} \Phi_r(\zeta)^{e_r}}.$$

If $e_s = 0$, the value is nonzero and the inverse is regular unless $G'_{n,k}(\zeta) = 0$. Any such derivative cancellation creates a higher ramification not explained by a cyclotomic zero.

This clean separation between *zero ramification* and *critical-value ramification* is useful computationally: cyclotomic points can be preclassified arithmetically before any numerical root search.

7.7 Exact low-degree examples

For $k = 1$, the Gaussian coefficient is the q -integer

$$G_{n,1}(q) = [n]_q = 1 + q + \cdots + q^{n-1}.$$

Thus $G_{2,1}^{-1}(Y) = Y - 1$, while

$$G_{3,1}^{-1}(Y) = \frac{-1 \pm \sqrt{4Y - 3}}{2}.$$

These examples already display the general principles: one global positive branch, square-root collisions at critical values, and multiple complex branches determined by polynomial degree.

Chapter 8

Continuous Gaussian coefficients and inversion in both parameters

Integer upper and lower indices cannot be varied analytically. The natural interpolation for $0 < q < 1$ is

$$\mathcal{C}_q(x, y) = \frac{\Gamma_q(x+1)}{\Gamma_q(y+1)\Gamma_q(x-y+1)}, \quad x > y > 0, \quad (8.1)$$

where

$$\Gamma_q(z) = (1-q)^{1-z} \frac{(q; q)_\infty}{(q^z; q)_\infty}. \quad (8.2)$$

At integral $x = n, y = k, \mathcal{C}_q(n, k) = \begin{bmatrix} n \\ k \end{bmatrix}_q$.

8.1 The $q \rightarrow 1$ bridge

A standard fixed-argument expansion, obtainable from Euler–Maclaurin or the asymptotics of Moak and McIntosh [13, 12], is

$$\begin{aligned} \log \Gamma_{e^{-t}}(z) = & \log \Gamma(z) - \frac{(z-1)(z-2)}{4}t \\ & + \frac{(z-2)(z-1)(2z+3)}{144}t^2 + O(t^3), \end{aligned} \quad (8.3)$$

locally uniformly for z in compact subsets avoiding nonpositive integers. Substitution in (8.1) gives the exact same first cumulants as the integer Gaussian polynomial.

Proposition 8.1 (Continuous Gaussian expansion at one). *Let*

$$d = y(x-y), \quad u = \log \frac{\mathcal{C}_{e^{-t}}(x, y)}{\binom{x}{y}}, \quad \binom{x}{y} = \frac{\Gamma(x+1)}{\Gamma(y+1)\Gamma(x-y+1)}.$$

Then

$$u = -\frac{d}{2}t + \frac{d(x+1)}{24}t^2 + O(t^3). \quad (8.4)$$

Consequently,

$$t = -\frac{2u}{d} + \frac{x+1}{3d^2}u^2 + O(u^3), \quad (8.5)$$

$$q = 1 + \frac{2u}{d} + \frac{5-x}{3d^2}u^2 + O(u^3). \quad (8.6)$$

Proof. Insert $z = x+1, y+1, x-y+1$ in (8.3). The linear combination of first coefficients is $-y(x-y)/2$, and the quadratic combination is $y(x-y)(x+1)/24$. Reversion gives the inverse. $\square$

8.2 Inverse in the upper parameter

Let $y > 0$ be fixed and regard $x \geq y$ as active. Write $\psi_q = (\log \Gamma_q)'$. From the Lambert series,

$$\psi_q(x) = -\log(1-q) + (\log q) \sum_{m \geq 0} \frac{q^{m+x}}{1-q^{m+x}}, \quad (8.7)$$

and $\psi'_q(x) > 0$.

Theorem 8.2 (Saturating upper-parameter inverse). *For fixed $y > 0$, $x \mapsto \mathcal{C}_q(x, y)$ is strictly increasing on $[y, \infty)$, with*

$$\mathcal{C}_q(y, y) = 1, \quad \lim_{x \rightarrow \infty} \mathcal{C}_q(x, y) = L_q(y) := \frac{(q^{y+1}; q)_\infty}{(q; q)_\infty} = \frac{1}{(1-q)^y \Gamma_q(y+1)}. \quad (8.8)$$

Thus it has a unique real inverse for $1 < Y < L_q(y)$.

Proof. Differentiate:

$$\partial_x \log \mathcal{C}_q(x, y) = \psi_q(x+1) - \psi_q(x-y+1) > 0.$$

At $x = y$, the value is one. Substituting (8.2) and letting $x \rightarrow \infty$ gives (8.8). $\square$

Near the lower endpoint, write $x = y + s$ and $u = \log \mathcal{C}_q(y + s, y)$. Then

$$u = \lambda_1 s + \frac{\lambda_2}{2} s^2 + \cdots, \quad \lambda_r = \psi_q^{(r-1)}(y+1) - \psi_q^{(r-1)}(1), \quad (8.9)$$

and $s(u)$ follows from (6.5) with λ_r in place of the order cumulants.

The upper endpoint has a logarithmic inverse. Put

$$r = q^{x-y+1}, \quad R = \frac{\mathcal{C}_q(x, y)}{L_q(y)}.$$

Then the exact expansion

$$\log R = - \sum_{m \geq 1} \frac{1 - q^{ym}}{m(1 - q^m)} r^m \quad (8.10)$$

shows, with $\Delta = 1 - R$ and $A = [y]_q = (1 - q^y)/(1 - q)$,

$$r = \frac{\Delta}{A} + \frac{A^2/2 - (1 - q^{2y})/[2(1 - q^2)]}{A^3} \Delta^2 + O(\Delta^3). \quad (8.11)$$

Hence

$$x = y - 1 + \frac{\log r}{\log q}, \quad (8.12)$$

which diverges logarithmically as $Y \uparrow L_q(y)$.

8.3 Inverse in the lower parameter

Fix $x > 0$ and let $0 < y < x$. Symmetry gives $\mathcal{C}_q(x, y) = \mathcal{C}_q(x, x - y)$. Set $g(y) = \log \mathcal{C}_q(x, y)$. Then

$$g'(y) = -\psi_q(y+1) + \psi_q(x-y+1), \quad (8.13)$$

$$g''(y) = -\psi'_q(y+1) - \psi'_q(x-y+1) < 0. \quad (8.14)$$

Theorem 8.3 (Two lower-parameter branches). *The function $y \mapsto \mathcal{C}_q(x, y)$ is strictly increasing on $(0, x/2)$, strictly decreasing on $(x/2, x)$, and has a unique maximum at $y = x/2$. Let*

$$u_0 = \frac{x}{2} + 1, \quad M = \mathcal{C}_q(x, x/2), \quad \delta = \log(M/Y).$$

For $Y \uparrow M$, the two inverse branches are

$$y = \frac{x}{2} \pm \sqrt{\frac{\delta}{\psi'_q(u_0)}} \left(1 - \frac{\psi_q'''(u_0)}{24\psi_q'(u_0)^2} \delta + O(\delta^2) \right). \quad (8.15)$$

Proof. The sign of (8.13) follows from strict increase of ψ_q . Around $y = x/2 + s$, symmetry removes odd powers and

$$\log M - g(x/2 + s) = \psi_q'(u_0)s^2 + \frac{\psi_q'''(u_0)}{12}s^4 + O(s^6).$$

Revert this even series and take the two square roots. $\square$

At the endpoint $y = 0$, $\mathcal{C}_q(x, 0) = 1$ and

$$\lambda_1 = \psi_q(x+1) - \psi_q(1) > 0, \quad \lambda_2 = -\psi_q'(1) - \psi_q'(x+1) < 0.$$

Thus, with $u = \log Y$,

$$y = \frac{u}{\lambda_1} - \frac{\lambda_2}{2\lambda_1^3}u^2 + O(u^3), \quad (8.16)$$

and the branch near $y = x$ follows by reflection.

8.4 Small and large base

The product form

$$\mathcal{C}_q(x, y) = \frac{(q^{x-y+1}; q)_\infty (q^{y+1}; q)_\infty}{(q^{x+1}; q)_\infty (q; q)_\infty} \quad (8.17)$$

shows that $\mathcal{C}_q(x, y) \rightarrow 1$ as $q \rightarrow 0^+$. Its expansion is a generalized power series whose exponent semigroup is generated by $1, y+1, x-y+1, x+1$. The universal leading term is

$$\mathcal{C}_q(x, y) = 1 + q + o(q),$$

because all three numerator/extra denominator exponents exceed one. Thus the positive inverse begins $q = Y - 1 + o(Y - 1)$, but fractional corrections appear when x or y is nonintegral.

For $q > 1$, define the interpolation by the standard reciprocity

$$\mathcal{C}_q(x, y) = q^{y(x-y)} \mathcal{C}_{1/q}(x, y). \quad (8.18)$$

It follows that every small- q generalized expansion has a corresponding large- q Laurent–Puisseux inverse with leading scale

$$q \sim Y^{1/[y(x-y)]}.$$

Conjecture 8.4 (Continuous positive-base monotonicity). *For every real $x > y > 0$, the map $q \mapsto \mathcal{C}_q(x, y)$ is strictly increasing on $(0, \infty)$ under the reciprocal continuation (8.18). This is automatic for integral (x, y) by coefficient positivity, and the expansions at $0, 1, \infty$ have the predicted sign, but a global proof for all real parameters appears to require a new positivity representation for $\partial_q \log \mathcal{C}_q$.*

Chapter 9

Inverse q -gamma, q -factorial, and q -beta functions

9.1 The inverse q -gamma function in its argument

For $0 < q < 1$, $\log \Gamma_q(x)$ is strictly convex on $(0, \infty)$ [3, 1]. The series (8.7) makes the geometry explicit.

Theorem 9.1 (Two real branches of Γ_q^{-1}). *There exists a unique $x_q > 0$ satisfying $\psi_q(x_q) = 0$. The number*

$$M_q = \Gamma_q(x_q)$$

is the unique global minimum. For every $Y > M_q$, the equation $\Gamma_q(x) = Y$ has exactly two positive real solutions, one in $(0, x_q)$ and one in (x_q, ∞) ; for $Y = M_q$ they coalesce.

Proof. As $x \downarrow 0$, the term $m = 0$ in (8.7) gives $\psi_q(x) = -1/x + O(1)$, so $\psi_q \rightarrow -\infty$. As $x \rightarrow \infty$, the sum tends to zero and $\psi_q(x) \rightarrow -\log(1 - q) > 0$. Strict convexity means $\psi'_q > 0$, so there is one zero. Also $\Gamma_q(x) \rightarrow \infty$ at both endpoints: near zero it has a simple pole, and at infinity the factor $(1 - q)^{1-x}$ grows exponentially. The branch count follows. $\square$

Let

$$p_j = \psi_q^{(j)}(x_q), \quad j \geq 1, \quad \delta = \log(Y/M_q).$$

Proposition 9.2 (Puisseux expansion at the minimum). *The two inverse branches have*

$$\begin{aligned} x = x_q \pm & \sqrt{\frac{2\delta}{p_1} - \frac{p_2}{3p_1^2}\delta} \\ & \pm \frac{\sqrt{2}(5p_2^2 - 3p_1p_3)}{36p_1^{7/2}}\delta^{3/2} + O(\delta^2). \end{aligned} \tag{9.1}$$

Proof. Taylor expand

$$\log \Gamma_q(x_q + s) - \log M_q = \frac{p_1}{2}s^2 + \frac{p_2}{6}s^3 + \frac{p_3}{24}s^4 + \dots$$

and substitute $a_2 = p_1/2, a_3 = p_2/6, a_4 = p_3/24$ in (2.6). $\square$

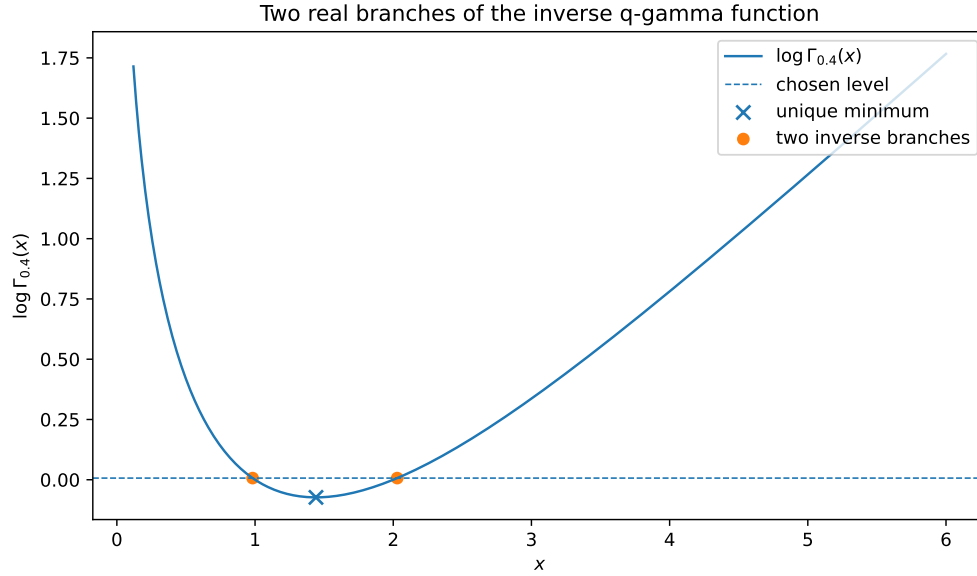


Figure 9.1: Strict log-convexity creates two real inverse q -gamma branches. They merge in a square-root singularity at the unique minimum.

9.2 Endpoint expansions of the two branches

Put $\ell = \log q < 0$ and

$$A_q = \frac{1-q}{-\ell}, \quad (9.2)$$

$$c_q = -\log(1-q) - \frac{1}{2}\ell + \ell \sum_{m \geq 1} \frac{q^m}{1-q^m}. \quad (9.3)$$

The Laurent expansion of the left branch is

$$\Gamma_q(x) = \frac{A_q}{x} + A_q c_q + O(x), \quad x \downarrow 0. \quad (9.4)$$

Hence

$$x = \frac{A_q}{Y} + \frac{A_q^2 c_q}{Y^2} + O(Y^{-3}) \quad (Y \rightarrow \infty) \quad (9.5)$$

for the left branch.

For the right branch, let

$$h = -\log(1-q) > 0, \quad X_0 = 1 + \frac{\log Y - \log(q; q)_\infty}{h}.$$

Since

$$-\log(q^x; q)_\infty = \frac{q^x}{1-q} + O(q^{2x}),$$

one obtains

$$x = X_0 - \frac{q^{X_0}}{h(1-q)} + O(q^{2X_0}). \quad (9.6)$$

Thus the two branches have qualitatively different endpoint scales: reciprocal on the left and affine-logarithmic with exponentially small correction on the right.

Since $[x]_q! = \Gamma_q(x+1)$, all results transfer immediately to the continuous q -factorial after shifting the active variable by one.

9.3 Inverse in the base of $\Gamma_q(x)$

A classical monotonicity theorem states that for $0 < q < r < 1$, $\Gamma_q(x) < \Gamma_r(x)$ when $0 < x < 1$ or $x > 2$, while the inequality is reversed for $1 < x < 2$; $\Gamma_q(1) = \Gamma_q(2) = 1$ [3, 10]. Consequently, for fixed $x \notin \{1, 2\}$, the equation $\Gamma_q(x) = Y$ has a unique $q \in (0, 1)$ whenever Y lies strictly between 1 and $\Gamma(x)$.

The exact small-base logarithm is

$$\log \Gamma_q(x) = (1-x) \log(1-q) - \sum_{m \geq 1} \frac{q^m - q^{mx}}{m(1-q^m)}. \quad (9.7)$$

It is a generalized power series in the semigroup generated by 1 and x . The leading inverse laws are

$$q \sim \begin{cases} (Y-1)^{1/x}, & 0 < x < 1, \\ \frac{Y-1}{x-2}, & 1 < x < 2 \text{ or } x > 2, \end{cases} \quad Y \rightarrow 1, \quad (9.8)$$

with the real side chosen according to the sign of $Y - 1$. The transition at $x = 1, 2$ is exact degeneracy, since $\Gamma_q(1) = \Gamma_q(2) = 1$ for all q .

For $q > 1$, the standard continuation satisfies

$$\Gamma_q(x) = q^{(x-1)(x-2)/2} \Gamma_{1/q}(x). \quad (9.9)$$

If $(x-1)(x-2) \neq 0$, inversion at $q = \infty$ therefore begins

$$q \sim Y^{2/[(x-1)(x-2)]}, \quad (9.10)$$

followed by generalized corrections inherited from (9.7). For negative or complex large q , a branch of q^x and $\text{Log } q$ must be fixed, and the inverse is sectorial.

9.4 Inverse q -beta functions

Define

$$B_q(a, b) = \frac{\Gamma_q(a) \Gamma_q(b)}{\Gamma_q(a+b)}, \quad a, b > 0.$$

For fixed b ,

$$\partial_a \log B_q(a, b) = \psi_q(a) - \psi_q(a+b) < 0,$$

so the active- a map is strictly decreasing.

Theorem 9.3 (One-parameter q -beta inverse). *For fixed $b > 0$, $a \mapsto B_q(a, b)$ maps $(0, \infty)$ bijectively onto*

$$((1-q)^b \Gamma_q(b), \infty). \quad (9.11)$$

As $Y \rightarrow \infty$,

$$a = \frac{A_q}{Y} + O(Y^{-2}), \quad (9.12)$$

where A_q is in (9.2). At the finite lower endpoint $L = (1-q)^b \Gamma_q(b)$, put $R = Y/L$ and $r = q^a$. Then

$$\log R = \sum_{m \geq 1} \frac{1 - q^{bm}}{m(1 - q^m)} r^m, \quad (9.13)$$

so

$$a = \frac{\log\{(R-1)/[b]_q\}}{\log q} + O(R-1). \quad (9.14)$$

Proof. Monotonicity was shown above. Near zero, the pole of $\Gamma_q(a)$ gives (9.12). At infinity, substitute the product formula for the ratio $\Gamma_q(a)/\Gamma_q(a+b)$, obtaining the limit and (9.13); apply endpoint saturation. $\square$

A second geometry appears if the sum $s = a + b$ is fixed. Then

$$H_s(a) = B_q(a, s-a), \quad 0 < a < s,$$

is symmetric and strictly log-convex because

$$(\log H_s)'' = \psi'_q(a) + \psi'_q(s-a) > 0.$$

It has a unique minimum at $a = s/2$, diverges at both endpoints, and its inverse has two square-root branches. If $\delta = \log(Y/H_s(s/2))$, then

$$a = \frac{s}{2} \pm \sqrt{\frac{\delta}{\psi'_q(s/2)}} \left(1 - \frac{\psi_q'''(s/2)}{24\psi_q'(s/2)^2} \delta + O(\delta^2) \right). \quad (9.15)$$

This is the convex dual of the concave lower-parameter geometry in [Theorem 8.3](#).

Chapter 10

Elementary q -analogues: exact and asymptotic inverses

The simplest examples are valuable because they expose, without the masking of a large special-function apparatus, almost every singular mechanism that recurs later: exact logarithmic inversion, algebraic base inversion, endpoint saturation, parity effects at $q = -1$, and a change from Taylor to Puiseux scales when an exponent crosses an integer threshold.

10.1 The q -number and its exact inverse in the exponent

For a fixed branch of $q^x = \exp(x \operatorname{Log} q)$, define

$$[x]_q = \frac{1 - q^x}{1 - q}, \quad q \neq 1, \quad [x]_1 = x. \quad (10.1)$$

The inverse with respect to x is elementary but multivalued.

Theorem 10.1 (Exact exponent inverse of the q -number). *Fix $q \in \mathbb{C} \setminus \{0, 1\}$, a branch of $\operatorname{Log} q$, and a branch of the logarithm at $1 - (1 - q)Y$. Every compatible local inverse branch is*

$$x = \frac{\operatorname{Log}(1 - (1 - q)Y) + 2\pi i \ell}{\operatorname{Log} q}, \quad \ell \in \mathbb{Z}. \quad (10.2)$$

For positive $q \neq 1$ and real levels for which the logarithm is real, the real branch is obtained by $\ell = 0$.

Proof. Equation $[x]_q = Y$ is equivalent to $q^x = 1 - (1 - q)Y$. Taking logarithms gives (10.2); conversely exponentiation recovers the level equation. $\square$

This formula is a useful model for parameterwise branch accounting. Changing ℓ changes the inverse sheet, while changing the branch of $\operatorname{Log} q$ changes the ambient definition of the forward function itself.

Put $\varepsilon = 1 - q$. Expansion of the exact inverse gives, uniformly for Y in compact subsets avoiding the logarithmic cut,

$$\begin{aligned} x = Y + \frac{Y(Y-1)}{2}\varepsilon + \frac{Y(Y-1)(4Y+1)}{12}\varepsilon^2 \\ + \frac{Y(Y-1)(6Y^2+2Y+1)}{24}\varepsilon^3 + O(\varepsilon^4). \end{aligned} \quad (10.3)$$

Thus the inverse tends to the identity as $q \rightarrow 1$, but its first correction is already nonlinear in the prescribed level.

The other endpoint regimes are singular. For fixed $Y \neq 1$,

$$x = \frac{\operatorname{Log}(1 - Y)}{\operatorname{Log} q} + O\left(\frac{q}{\operatorname{Log} q}\right), \quad q \rightarrow 0, \quad (10.4)$$

so all finite nonunit levels are compressed toward $x = 0$. As $q \rightarrow \infty$ in a fixed sector,

$$x = 1 + \frac{\text{Log } Y}{\text{Log } q} + O\left(\frac{1}{q \text{Log } q}\right). \quad (10.5)$$

The points $x = 0$ and $x = 1$ are therefore the two endpoint condensers for base degeneration.

10.2 Inverse in the base for integral exponent

For $n \in \mathbb{N}$, $n \geq 2$,

$$[n]_q = 1 + q + \cdots + q^{n-1} \quad (10.6)$$

is a polynomial of degree $n - 1$. Its inverse is algebraic, with branch points at the critical values of this polynomial.

10.2.1 The branch at $q = 0$

Write $\delta = Y - 1$. For $n \geq 6$, the inverse through fifth order is stable with respect to n :

$$q = \delta - \delta^2 + \delta^3 - \delta^4 + \delta^5 + O(\delta^6). \quad (10.7)$$

More generally, the coefficients agree with the expansion of $\delta/(1 + \delta)$ through degree $n - 1$. This follows from $[n]_q = (1 - q^n)/(1 - q)$: the term q^n cannot influence reversion below order n .

For nonintegral x , the same base inverse becomes a generalized Puiseux problem. If $0 < \Re x < 1$, then

$$[x]_q = 1 - q^x + q + \cdots, \quad q \sim (1 - Y)^{1/x}. \quad (10.8)$$

If $\Re x > 1$, the ordinary term q leads and $q \sim Y - 1$, but the correction q^x creates a nonintegral transseries whenever $x \notin \mathbb{N}$. This elementary example explains why a uniform “Taylor series in q ” is not available for all continuous parameters.

10.2.2 The branch at $q = 1$

Let $h = q - 1$, $\delta = Y - n$, and

$$A = \binom{n}{2}, \quad B = \binom{n}{3}, \quad C = \binom{n}{4}, \quad D = \binom{n}{5}. \quad (10.9)$$

Since $[n]_{1+h} = n + Ah + Bh^2 + Ch^3 + Dh^4 + \cdots$, reversion yields

$$\begin{aligned} q = 1 + & \frac{\delta}{A} - \frac{B}{A^3}\delta^2 + \frac{2B^2 - AC}{A^5}\delta^3 \\ & + \frac{-5B^3 + 5ABC - A^2D}{A^7}\delta^4 + O(\delta^5). \end{aligned} \quad (10.10)$$

This is the elementary prototype of the Gaussian-coefficient formula in [Chapter 7](#).

10.2.3 The parity point $q = -1$

For $n = 2m$ and $n = 2m + 1$, respectively,

$$[2m]_{-1} = 0, \quad [2m]'_{-1} = m; \quad [2m + 1]_{-1} = 1, \quad [2m + 1]'_{-1} = -m. \quad (10.11)$$

Consequently the local inverses are regular for $m \geq 1$:

$$q = -1 + \begin{cases} Y/m + O(Y^2), & n = 2m, \\ -(Y - 1)/m + O((Y - 1)^2), & n = 2m + 1. \end{cases} \quad (10.12)$$

The analogous point for Gaussian coefficients is subtler because cyclotomic multiplicity can make the derivative vanish.

10.2.4 The branch at infinity

Put $d = n - 1$ and $Z = Y^{1/d}$. For $n \geq 3$, on each chosen d -th root branch,

$$q = Z - \frac{1}{d} - \frac{d+1}{2d^2}Z^{-1} + O(Z^{-2}). \quad (10.13)$$

For $n = 2$ the exact answer is $q = Y - 1$. Negative infinity is obtained by the appropriate root branch; when d is even, a large positive level has two real tails, whereas when d is odd it has one.

10.3 Jackson's two q -exponentials and their inverses

For $|q| < 1$, let

$$e_q(z) = \frac{1}{(z; q)_\infty}, \quad E_q(z) = (-z; q)_\infty. \quad (10.14)$$

Their compositional inverses in z are not new transcendental objects independent of the Pochhammer inverse; exactly,

$$e_q^{-1}(Y) = \mathcal{B}_{a,q}(Y^{-1}), \quad E_q^{-1}(Y) = -\mathcal{B}_{a,q}(Y), \quad (10.15)$$

where $\mathcal{B}_{a,q}$ denotes a chosen inverse branch of $a \mapsto (a; q)_\infty$.

At $q = 0$ the inverses collapse to elementary functions:

$$e_0(z) = \frac{1}{1-z}, \quad e_0^{-1}(Y) = 1 - Y^{-1}; \quad E_0(z) = 1 + z, \quad E_0^{-1}(Y) = Y - 1. \quad (10.16)$$

At fixed q , it is often better to invert the logarithmic level $L = \text{Log } Y$. One has

$$\text{Log } e_q(z) = \sum_{m \geq 1} \frac{z^m}{m(1-q^m)}, \quad (10.17)$$

$$\text{Log } E_q(z) = \sum_{m \geq 1} \frac{(-1)^{m+1} z^m}{m(1-q^m)}. \quad (10.18)$$

The universal formula (2.1) therefore gives all local coefficients at $z = 0$.

The classical limit is regular only after scaling the argument. Set $q = 1 - \varepsilon$, and define z_e and z_E by

$$e_q(\varepsilon z_e) = Y, \quad E_q(\varepsilon z_E) = Y, \quad L = \text{Log } Y. \quad (10.19)$$

Direct inversion of (10.17)–(10.18) gives

$$z_e = L - \frac{L^2}{4}\varepsilon + \frac{L^2(L-9)}{72}\varepsilon^2 - \frac{L^2(L^2-8L+36)}{576}\varepsilon^3 + O(\varepsilon^4), \quad (10.20)$$

$$z_E = L + \frac{L^2}{4}\varepsilon + \frac{L^2(L+9)}{72}\varepsilon^2 + \frac{L^2(L^2+8L+36)}{576}\varepsilon^3 + O(\varepsilon^4). \quad (10.21)$$

Both tend to the ordinary logarithm. Their odd corrections have opposite sign, reflecting the identity $e_q(z)E_q(-z) = 1$.

Remark 10.2 (Base inversion). Solving $e_q(z) = Y$ or $E_q(z) = Y$ for q is precisely the base inverse of the infinite Pochhammer symbol. Hence the radial root-of-unity expansion, including $q \rightarrow -1$, transfers without further work after replacing the level by Y^{-1} or replacing a by $-z$. No canonical $q \rightarrow \infty$ statement exists for the product definitions (10.14) until a continuation outside the unit disk has been chosen.

10.4 A normalized q -polylogarithm family

The inverse theory of a whole class of Lambert-series analogues can be presented at once. For $0 < q < 1$, $s \in \mathbb{C}$, and $|z| < 1$, define

$$\mathcal{L}_{s,q}(z) = \sum_{m \geq 1} \frac{z^m}{[m]_q^s}. \quad (10.22)$$

At $q = 1$ this is $\text{Li}_s(z)$; at $q = 0$ it is $z/(1 - z)$ for every s . The case $s = 1$ is a normalized Jackson-type logarithm.

Proposition 10.3 (Local inverse in z). *Put $A_j = [j]_q^{-s}$. The branch satisfying $z(0) = 0$ is*

$$\begin{aligned} z = Y - A_2 Y^2 + (2A_2^2 - A_3)Y^3 + (-5A_2^3 + 5A_2 A_3 - A_4)Y^4 \\ + (14A_2^4 - 21A_2^2 A_3 + 6A_2 A_4 + 3A_3^2 - A_5)Y^5 + O(Y^6). \end{aligned} \quad (10.23)$$

At the exact endpoint $q = 0$, this sums to $z = Y/(1 + Y)$.

Proof. Apply fifth-order reversion to $Y = z + A_2 z^2 + A_3 z^3 + \dots$. At $q = 0$, all $A_j = 1$, so the forward series is geometric. $\square$

The perturbation from the classical polylogarithm is especially simple. Since

$$[m]_{1-\varepsilon} = m - \binom{m}{2} \varepsilon + O(\varepsilon^2), \quad (10.24)$$

we obtain

$$\mathcal{L}_{s,1-\varepsilon}(z) = \text{Li}_s(z) + \frac{s\varepsilon}{2} (\text{Li}_{s-1}(z) - \text{Li}_s(z)) + O(\varepsilon^2). \quad (10.25)$$

If z_0 is a regular branch of $\text{Li}_s^{-1}(Y)$, then

$$z = z_0 - \frac{s\varepsilon z_0}{2} \left(1 - \frac{Y}{\text{Li}_{s-1}(z_0)} \right) + O(\varepsilon^2). \quad (10.26)$$

For $s = 1$, $z_0 = 1 - e^{-Y}$, giving a fully elementary classical anchor.

There are also useful global parameter inverses.

Theorem 10.4 (Inverse in q and in s). *Let $0 < z < 1$.*

- (i) *If $s > 0$, then $q \mapsto \mathcal{L}_{s,q}(z)$ is strictly decreasing on $(0, 1)$, from $z/(1 - z)$ to $\text{Li}_s(z)$. Hence each intermediate level has a unique base inverse.*
- (ii) *If $0 < q < 1$, then $s \mapsto \mathcal{L}_{s,q}(z)$ is strictly decreasing on $\mathbb{R}$, and tends to z as $s \rightarrow +\infty$. Its upper endpoint inverse satisfies*

$$s = \frac{\log\{z^2/(Y - z)\}}{\log[2]_q} + o(1), \quad Y \downarrow z. \quad (10.27)$$

Proof. For $m \geq 2$, $[m]_q$ increases strictly with q , proving (i) termwise. Differentiating in s gives

$$\partial_s \mathcal{L}_{s,q}(z) = - \sum_{m \geq 2} \frac{z^m \log[m]_q}{[m]_q^s} < 0.$$

As $s \rightarrow \infty$, the $m = 1$ term survives and the $m = 2$ term is the first correction: $Y - z = z^2 [2]_q^{-s} (1 + o(1))$. Taking logarithms proves (10.27). $\square$

Near the base endpoint $q = 0$, for fixed s ,

$$\mathcal{L}_{s,q}(z) = \frac{z}{1-z} - \frac{sz^2}{1-z}q + O(q^2), \quad (10.28)$$

locally uniformly for s in compact sets. Hence, for $s \neq 0$, the local real inverse has leading term

$$q = \frac{z - (1-z)Y}{sz^2} + o\left(\frac{z}{1-z} - Y\right). \quad (10.29)$$

For fixed s , this is an ordinary analytic germ at $q = 0$. Fractional powers can arise only in a double scaling in which s itself depends singularly on q , not in the fixed-parameter expansion.

Chapter 11

Basic hypergeometric functions and parameterwise inversion

The Pochhammer and Gaussian cases are the atoms of basic hypergeometric series. Once their logarithmic derivatives and local inverses are known, a large class of parameter inversions becomes algorithmic. We use the standard normalization of [6, 1].

11.1 The inverse in the series variable

Let

$${}_r\phi_s\left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; q, z\right) = \sum_{m \geq 0} c_m z^m, \quad (11.1)$$

where

$$c_m = \frac{(a_1, \dots, a_r; q)_m}{(q, b_1, \dots, b_s; q)_m} \left((-1)^m q^{\binom{m}{2}}\right)^{1+s-r}. \quad (11.2)$$

Assume no denominator factor vanishes and $c_1 \neq 0$.

Theorem 11.1 (Universal local inverse of a basic hypergeometric series). *For $\delta = Y - 1$, the branch through $z = 0$ begins*

$$\begin{aligned} z = & \frac{\delta}{c_1} - \frac{c_2}{c_1^3} \delta^2 + \frac{2c_2^2 - c_1 c_3}{c_1^5} \delta^3 \\ & + \frac{-5c_2^3 + 5c_1 c_2 c_3 - c_1^2 c_4}{c_1^7} \delta^4 + O(\delta^5). \end{aligned} \quad (11.3)$$

The complete coefficient sequence is given by (2.1). If $c_1 = \dots = c_{d-1} = 0$ and $c_d \neq 0$, the inverse is instead a d -sheeted Puiseux germ $z \sim (\delta/c_d)^{1/d}$.

Proof. This is ordinary series reversion applied to $Y - 1 = \sum_{m \geq 1} c_m z^m$. The degenerate statement follows from Theorem 2.3. $\square$

The theorem includes terminating q -orthogonal polynomials after an affine normalization of the dependent variable. In a terminating case $a_i = q^{-N}$, the forward map is a polynomial of degree at most N in z , and the complete inverse surface is algebraic. Its branch values are the zeros of the discriminant of the level polynomial.

11.2 Derivatives with respect to every continuous parameter

For a finite Pochhammer factor away from its zeros,

$$\partial_a \log(a; q)_m = - \sum_{j=0}^{m-1} \frac{q^j}{1 - aq^j}, \quad (11.4)$$

$$\partial_q \log(a; q)_m = - \sum_{j=1}^{m-1} \frac{ajq^{j-1}}{1 - aq^j}. \quad (11.5)$$

Consequently, termwise differentiation of (11.1) gives, for an upper parameter a_i ,

$$\partial_{a_i} F = - \sum_{m \geq 1} c_m z^m \sum_{j=0}^{m-1} \frac{q^j}{1 - a_i q^j}, \quad (11.6)$$

and for a lower parameter b_i ,

$$\partial_{b_i} F = \sum_{m \geq 1} c_m z^m \sum_{j=0}^{m-1} \frac{q^j}{1 - b_i q^j}. \quad (11.7)$$

The base derivative is obtained by combining (11.5) for every factor with the derivative of the explicit power of q in (11.2).

Corollary 11.2 (Parameter sensitivity of the inverse). *Let $F(z; \theta) = Y$, with $F_z \neq 0$. For every continuous upper or lower parameter, and for the base away from its singular set,*

$$\frac{\partial z}{\partial \theta} = - \frac{F_\theta}{F_z}. \quad (11.8)$$

If the level also varies,

$$dz = \frac{dY - \sum_j F_{\theta_j} d\theta_j}{F_z}. \quad (11.9)$$

Equation (11.9) is the differential equation for continuing a chosen inverse sheet through parameter space. It also identifies the numerical condition number: every inverse becomes ill-conditioned when F_z approaches zero.

11.3 Terminating parameters and singular lower parameters

An upper parameter $a_i = q^{-N}$ truncates the series because $(q^{-N}; q)_m = 0$ for $m > N$. Varying a_i away from this value reactivates the tail. Inside any compact parameter subdomain on which the nonterminating series converges normally and no denominator factor vanishes, the variation is analytic. Thus a terminating point is not automatically a branch point of the parameter inverse. It becomes one only if the corresponding parameter derivative (11.6) vanishes.

A lower parameter $b_i = q^{-N}$, by contrast, lies on the polar divisor of the generic nonterminating function. Near a simple polar factor the forward map has the form

$$F(z; b_i) = \frac{A(z)}{1 - b_i q^N} + B(z) + O(1 - b_i q^N), \quad (11.10)$$

with $A(z) \neq 0$ generically. Inverting a large level gives the reciprocal scale

$$b_i = q^{-N} \left(1 - \frac{A(z)}{Y} + O(Y^{-2}) \right). \quad (11.11)$$

When numerator and denominator zeros cancel, one must first remove the singularity; the inverse type is determined by the first nonvanishing term of the reduced germ, not by the unreduced quotient.

11.4 Perturbative inversion near $q = 1$

A broad class of scaled basic hypergeometric functions has an expansion

$$F_q(z) = F_0(z) + \varepsilon F_1(z) + \varepsilon^2 F_2(z) + \cdots, \quad q = 1 - \varepsilon, \quad (11.12)$$

where F_0 is an ordinary hypergeometric or confluent hypergeometric function. The correct scaling of z depends on $r - s$, so no single unscaled formula covers all ${}_r\phi_s$.

Proposition 11.3 (Inverse perturbation about the classical limit). *Let z_0 satisfy $F_0(z_0) = Y$ and $F'_0(z_0) \neq 0$. The corresponding inverse branch of (11.12) is*

$$z = z_0 - \varepsilon \frac{F_1(z_0)}{F'_0(z_0)} + \varepsilon^2 \left[-\frac{F_2(z_0)}{F'_0(z_0)} + \frac{F_1(z_0)F'_1(z_0)}{F'_0(z_0)^2} - \frac{F''_0(z_0)F_1(z_0)^2}{2F'_0(z_0)^3} \right] + O(\varepsilon^3). \quad (11.13)$$

Proof. Substitute $z = z_0 + \varepsilon z_1 + \varepsilon^2 z_2 + \cdots$ into (11.12), Taylor expand, and equate powers of ε . $\square$

This proposition turns any known forward $q \rightarrow 1$ expansion into an inverse expansion. The true obstruction is not reversion but uniformity: if $F'_0(z_0) = 0$, the expansion must be rescaled and replaced by the Puiseux normal form of [Theorem 2.3](#).

11.5 Roots of unity and base singularities

At a root of unity, denominator factors $(q; q)_m$ and $(b_i; q)_m$ can vanish periodically. There are three distinct cases.

- (a) A terminating numerator kills every potentially singular term. The function has a finite removable limit, and ordinary or Puiseux inversion applies to the reduced polynomial.
- (b) A finite number of uncanceled terms is singular. The parameter inverse has an algebraic reciprocal scale such as (11.11).
- (c) Infinitely many terms are coherently singular. The radial asymptotic is usually exponential in $1/t$, $q = \zeta e^{-t}$, and inversion takes the logarithmic form $t \asymp A/\text{Log } Y$. Cyclic quantum dilogarithms and root-of-unity Pochhammer asymptotics provide the forward coefficients in this case; see [11, 4] for related modern contexts.

Thus a root of unity is not one universal singularity. Cancellation data and the order of termination decide the inverse scale.

11.6 Orthogonal-polynomial and q -Appell specializations

Many families in the q -Askey scheme are terminating ${}_r\phi_s$ functions [8]. For a fixed degree n , inversion in the polynomial variable is therefore algebraic. A useful uniform statement is the following.

Proposition 11.4 (Fixed-degree inverse atlas). *Let $P_n(x; q, \theta)$ be a degree- n polynomial in x , analytic in continuous parameters on a domain U , with nonzero leading coefficient. Then the multivalued inverse of $P_n(x; q, \theta) = Y$ is an n -sheeted branched covering of the (Y, q, θ) -space. Its finite branch divisor is contained in*

$$\text{Disc}_x(P_n(x; q, \theta) - Y) = 0, \quad (11.14)$$

and its local expansions are analytic at simple points and Puiseux at points of higher ramification.

For $q \rightarrow 1$, the branches may be continued from the inverse branches of the corresponding classical orthogonal polynomial, except where classical critical values collide. For $n \rightarrow \infty$, however, the fixed-degree algebraic atlas is not uniform; turning points and edge scalings are expected to generate Airy-type inverse transitions.

Chapter 12

Simultaneous inversion, condition numbers, and branch geometry

A scalar equation cannot determine several unknown parameters without additional constraints. The phrase “inverse with respect to all parameters” therefore means, first, inverting one parameter at a time while holding the others fixed; and, second, solving a square system of independent level equations when several parameters are to be recovered simultaneously.

12.1 Iso-level hypersurfaces

For one scalar observable $F(\theta_1, \dots, \theta_p) = Y$, a regular level is a hypersurface. If $F_{\theta_p} \neq 0$, the local graph $\theta_p = \mathcal{B}(Y, \theta_1, \dots, \theta_{p-1})$ satisfies

$$\partial_Y \mathcal{B} = \frac{1}{F_{\theta_p}}, \quad \partial_{\theta_j} \mathcal{B} = -\frac{F_{\theta_j}}{F_{\theta_p}}. \quad (12.1)$$

For $(a; q)_\alpha$, this gives a three-directional inverse calculus in a, q, α . The vanishing of one chosen denominator derivative does not make the level hypersurface singular; it means only that the selected graph coordinate has failed. Another parameter may remain a valid local graph.

12.2 Several observables and the inverse Jacobian

Let $\mathbf{F} = (F_1, \dots, F_p)$ and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_p)$. If

$$J(\boldsymbol{\theta}) = \det \left(\frac{\partial F_i}{\partial \theta_j} \right)_{i,j=1}^p \neq 0, \quad (12.2)$$

then $\boldsymbol{\theta}$ is locally an analytic function of the level vector $\mathbf{Y}$, and

$$d\boldsymbol{\theta} = (D\mathbf{F})^{-1} d\mathbf{Y}. \quad (12.3)$$

For example, two Pochhammer observations

$$(a; q)_{n_1} = Y_1, \quad (a; q)_{n_2} = Y_2 \quad (12.4)$$

can locally recover both a and q whenever the two logarithmic-gradient rows are independent. This is a natural inverse-identification problem for experimental or symbolic data.

12.3 Relative condition numbers

For a scalar inverse $F(z) = Y$, define the local relative condition number

$$\kappa_{z \leftarrow Y} = \left| \frac{Y}{zF'(z)} \right|, \quad (12.5)$$

when $zY \neq 0$. At a critical point of multiplicity $m - 1$, $z - z_c \asymp (Y - Y_c)^{1/m}$, so the absolute derivative of the inverse diverges like $|Y - Y_c|^{1/m-1}$. Near an endpoint saturation $Y - L \asymp q^\alpha$, the inverse in α has derivative

$$\frac{d\alpha}{dY} \asymp \frac{1}{(Y - L) \log q}, \quad (12.6)$$

which is logarithmically severe. These estimates dictate both working precision and continuation step size.

12.4 Monodromy as the global inverse invariant

For a finite polynomial level equation, analytic continuation of inverse branches around critical values permutes its sheets. The resulting monodromy group is a finer invariant than the discriminant alone. In the finite Pochhammer case, the generic polynomial in a has prescribed geometrically spaced zeros, but its critical values are not geometrically spaced. In the base variable the exponents form a sparse subset-sum pattern, and the monodromy can change on algebraic loci in a .

For infinite products, branch points accumulate through zero sets and radial boundary singularities. The appropriate global object is no longer a finite branched covering but a Riemann surface with logarithmic ends. The root-of-unity expansions in [Section 3.5](#) and the radial formulas for $(a; q)_\infty$ are local coordinate charts on this surface.

Chapter 13

Branch-aware numerical inversion and experiments

13.1 Why direct Newton iteration is not enough

A black-box solve of $F(z) = Y$ can converge to the wrong sheet, cross a product zero, or lose all significant digits near a critical level. The analytic work above supplies three pieces of information a robust solver must use:

- (i) a local coordinate and asymptotic seed;
- (ii) a branch label, usually inherited by continuation from a known point;
- (iii) a representation of F , F' , and often F'' that remains stable in the selected regime.

For products with very large or very small magnitude, solve the logarithmic equation

$$G(z) = \text{Log } F(z) - \text{Log } Y = 0 \quad (13.1)$$

instead of subtracting two nearly equal huge numbers. The branch of the logarithm must be continued consistently.

13.2 Newton, Halley, and continuation equations

Newton's step for (13.1) is

$$z_{r+1} = z_r - \frac{G(z_r)}{G'(z_r)}. \quad (13.2)$$

When a stable second derivative is available, Halley's step is

$$z_{r+1} = z_r - \frac{2G(z_r)G'(z_r)}{2G'(z_r)^2 - G(z_r)G''(z_r)}. \quad (13.3)$$

Near a simple solution this is cubically convergent. Near a ramification point, neither method should be applied in the original coordinate; use $w = (Y - Y_c)^{1/m}$ or factor out the known multiplicity first.

To track a sheet as a passive parameter s changes, integrate

$$\frac{dz}{ds} = -\frac{F_s(z; s)}{F_z(z; s)}. \quad (13.4)$$

Predictor–corrector continuation combines an ODE step with one or two Newton corrections. A small loop in parameter space can be used to compute the monodromy permutation numerically.

Algorithm 13.1 (Branch-aware inverse solver). Given a level Y , parameters, and a desired branch:

1. Identify the nearest regular point, zero, critical point, endpoint, root of unity, or infinite regime.
2. Select the corresponding coordinate: $Y - Y_0$, a fractional power, $-\text{Log } q$, $1/q$, q^α , or $1/\text{Log } Y$.
3. Evaluate the derived asymptotic expansion to obtain a branch-labelled seed.
4. Evaluate products through logarithmic sums and Gaussian polynomials through recurrence when quotient formulas contain removable singularities.
5. Correct with safeguarded Newton or Halley iteration. On a real monotone interval, retain a bracketing interval.
6. Verify the residual and, when possible, a posteriori certify uniqueness with an interval bound on F' .

13.3 Numerical branch portraits

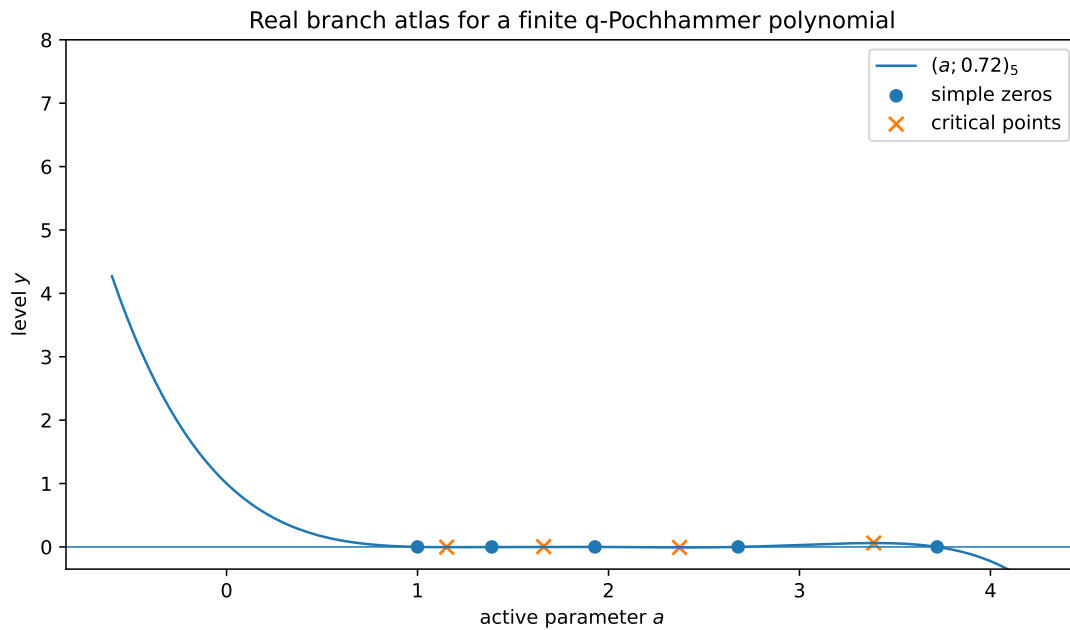


Figure 13.1: A real branch atlas for a finite Pochhammer level map. Critical points partition the real axis into monotonicity intervals; a horizontal level may have several real inverse values.

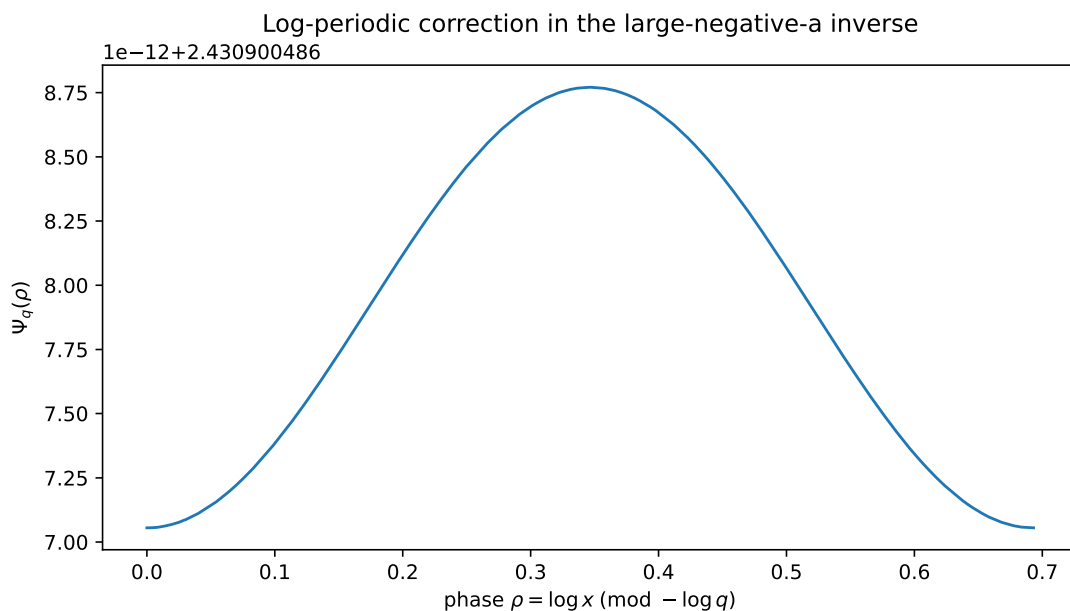


Figure 13.2: Bounded log-periodic correction in the large-negative- a asymptotic of $(a; q)_\infty$. The period in $\log(-a)$ is $|\log q|$, so the inverse carries a small periodic modulation on top of its square-root logarithmic scale.

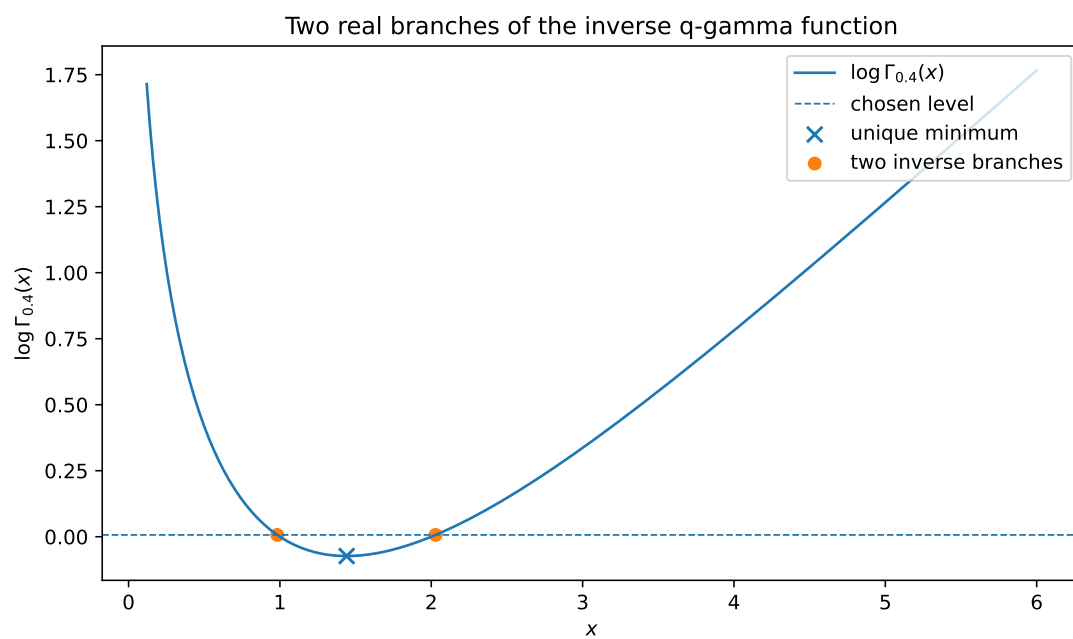


Figure 13.3: The two real inverse branches of $\Gamma_q(x) = Y$ for a fixed $0 < q < 1$. They meet at the unique minimum in a square-root fold.

13.4 High-precision checks

The accompanying program uses 80 decimal digits, logarithmic summation for infinite products, a q -Pascal recurrence for Gaussian coefficients, and secant correction from the asymptotic seeds. Representative checks are in Table 13.1. The errors decrease at the predicted powers when the displacement is reduced; the table records one scale per regime.

Table 13.1: Representative high-precision checks of the inverse expansions. The final column is absolute error.

Experiment	exact	asymptotic	error
$(a; q)_7, a \leftarrow y, q = 0.37$	0.012	0.012	$7.551e - 13$
$(a; q)_8$, branch at $a = q^{-3}$	4.4056553	4.4056553	$7.806e - 21$
$(0.42; q)_9, q \leftarrow y, q \sim 0$	0.025	0.025000004	$4.19e - 9$
$(0.31; q)_8, q \leftarrow y, q \sim 1$	0.975	0.9750017	$1.701e - 6$
$(0.30; q)_\infty$, radial $q \rightarrow 1$	0.96560542	0.96560542	$2.699e - 13$
$(0.30; q)_\infty$, radial $q \rightarrow -1$	-0.96560542	-0.96560543	$1.775e - 8$
$\begin{bmatrix} 13 \\ 5 \end{bmatrix}_q, q \sim 0$	0.018	0.018000008	$7.824e - 9$
$\begin{bmatrix} 12 \\ 5 \end{bmatrix}_q, q \sim 1$	0.985	0.98500014	$1.427e - 7$
$\begin{bmatrix} 12 \\ 5 \end{bmatrix}_q$, simple zero at $q = -1$	-0.9999998	-0.9999998	$6.8e - 13$
$\Gamma_{0.4}^{-1}$, left minimum branch	1.432631	1.432631	$1.959e - 11$
$\Gamma_{0.4}^{-1}$, right minimum branch	1.4490496	1.4490496	$1.961e - 11$
$C_{0.5}(7.3, y)^{-1}$, left center branch	3.6277547	3.6277547	$7.035e - 12$
$C_{0.5}(7.3, y)^{-1}$, right center branch	3.6722453	3.6722453	$7.035e - 12$
$e_q((1 - q)z)^{-1}, q = 0.97$	0.8	0.80000001	$8.978e - 9$

The experiment at $q = -1$ deliberately evaluates the Gaussian polynomial by recurrence. A product quotient would present removable $0/0$ factors and is numerically unsuitable. Likewise, the $q \rightarrow \pm 1$ infinite-product experiments sum logarithms until a precision-dependent tail bound is met.

13.5 Reproducibility

The archive contains `inverse_q_analogs_experiments.py`. Running

```
python inverse_q_analogs_experiments.py
```

regenerates `numerical_results.tex`, a plain-text log, and the three figures. The script contains no network calls and records all parameter choices next to the corresponding check. A complete run requires Python, `mpmath`, `sympy`, and `matplotlib`.

Chapter 14

Conjectures and frontier problems

The following statements are intentionally separated from the proved results. They are designed to be testable through exact computation, validated numerics, or a sharply defined analytic program.

14.1 Generic critical geometry of finite Pochhammer maps

Conjecture 14.1 (Generic simplicity in the base). *Fix $n \geq 3$. Outside a proper algebraic subset of $a \in \mathbb{C}$, every finite critical point of $q \mapsto (a; q)_n$ is simple and distinct critical points have distinct critical values. Equivalently, the level polynomial has only simple ramification away from the explicit collision loci forced by $a = 0, 1$ and roots of unity.*

The conjecture can be attacked by computing two resultants: one detecting a multiple zero of $\partial_q(a; q)_n$, and one detecting equality of two critical values. It is enough to exhibit a single parameter value outside each resultant locus to prove that the exceptional set is proper for every fixed n . A uniform proof in n would likely exploit the sparse subset-sum support of the polynomial.

Problem 14.2 (Monodromy groups). *Determine the generic monodromy group of the inverse in a and in q for $(a; q)_n = Y$. Decide when the geometric progression of roots forces a proper subgroup of the full symmetric group.*

14.2 Real and sectorial univalence

The real critical points of $a \mapsto (a; q)_n$ for $0 < q < 1$ interlace the positive zeros. A complex analogue should control inverse branches in sectors.

Conjecture 14.3 (Complex thickening of real inverse sheets). *For fixed $0 < q < 1$, each real monotonicity interval I determined by the zeros and critical points of $P_n(a) = (a; q)_n$ is contained in a maximal simply connected domain D_I , symmetric about the real axis, on which P_n is univalent. The boundary of D_I is assembled from critical trajectories issuing from zeros and critical points. As $n \rightarrow \infty$, domains whose real cores remain in a fixed compact set converge in the Carathéodory sense to univalence domains of $(a; q)_\infty$.*

A practical first step is to seek domains on which a rotation of P'_n has positive real part, so that the Noshiro–Warschawski criterion applies, or to establish starlikeness through bounds for $aP'_n(a)/P_n(a)$. Numerical critical trajectories can be used to formulate the correct boundary arcs before attempting a uniform proof.

14.3 Root-of-unity transseries and monodromy

Conjecture 14.4 (All-order radial inverse at a root of unity). *Let ζ be a primitive d -th root of unity and let a avoid the radial singular set. The inverse of $(a; \zeta e^{-t})_\infty = Y$ admits, in each non-Stokes sector, a*

complete transseries in odd and even powers of X^{-1} ,

$$t \sim \frac{A_d(a)}{X} \left(1 + \sum_{m \geq 1} \frac{P_m(a, \zeta)}{X^m} \right), \quad X = -\operatorname{Log} Y + C_d(a, \zeta), \quad (14.1)$$

whose coefficients lie in the differential field generated by polylogarithms of a^d and finite cyclic dilogarithmic sums. Stokes jumps act by explicit translations of X and generate the local inverse monodromy.

Recent work on root-of-unity asymptotics, quantum dilogarithms, and modular resurgence shows that the forward expansions carry nontrivial exponentially small data [11, 4]. The frontier question is how that data transforms under compositional inversion.

14.4 Large-argument log-periodic inversion

Conjecture 14.5 (Full log-periodic inverse expansion). *Let $0 < q < 1$, $\tau = |\log q|$, $a = -e^L$, and $R = \sqrt{2\tau \operatorname{Log} Y}$. The asymptotic inverse of $(a; q)_\infty = Y$ has a complete expansion*

$$L = R - \frac{\tau}{2} + \sum_{m \geq 1} \frac{C_m(\theta)}{R^m}, \quad \theta = \left\{ \frac{R - \tau/2}{\tau} \right\}, \quad (14.2)$$

where the C_m are periodic and smooth away from a finite set of matching phases. The first coefficient is predicted by the proved split-product formula to be

$$C_1(\theta) = \tau \left(\frac{\tau}{8} - \Psi_q(\tau\theta) \right).$$

Exponentially small corrections should complete the algebraic expansion to a sectorial transseries.

A Mellin–Poisson treatment should generate the higher periodic coefficients and determine whether phase matching produces exponentially small Stokes switches.

14.5 Inverse Gaussian coefficients and partition coefficients

Problem 14.6 (Asymptotics of inverse partition coefficients). *In a stable rectangle, write*

$$q = \sum_{m \geq 1} d_m \left(\left[\begin{matrix} n \\ k \end{matrix} \right]_q - 1 \right)^m. \quad (14.3)$$

Determine the sign pattern, integrality properties, exact combinatorial model, and asymptotic growth of d_m . In the unrestricted stable range the first terms are

$$1, -2, 5, -15, 52, \dots \quad (14.4)$$

Is there a sign-reversing model on rooted partition structures that explains these coefficients directly?

Problem 14.7 (Global proof of continuous Gaussian monotonicity). *Prove or disprove Theorem 8.4. A proof would turn the local base inverses of $\mathcal{C}_q(x, y)$ into global real branches. The logarithmic derivative is an explicit Lambert series, but its sign involves a nontrivial cancellation among the three q -gamma terms.*

14.6 Differential and q -difference transcendence

Conjecture 14.8 (Generic nonholonomicity of compositional inverses). *Apart from algebraic and explicitly logarithmic degeneracies, a generic local compositional inverse of $(a; q)_\infty$, $\Gamma_q(x)$, or a nonterminating ${}_r\phi_s$ is neither D -finite in its level nor q -holonomic with respect to that level.*

The forward functions satisfy rich q -difference identities, but compositionally inverting a nonlinear function rarely preserves linear differential or recurrence structure. A possible route is to combine singularity proliferation of the inverse surface with closure theorems for D -finite functions. Exceptional parameter values should be classified rather than excluded silently.

14.7 Uniform double scaling at coalescing branches

The local formulas in this report assume one distinguished singular mechanism at a time. Several important limits violate that separation:

- (i) $q \rightarrow 1$ while the target approaches a critical value;
- (ii) $q \rightarrow \zeta$ while an active parameter approaches a colliding zero;
- (iii) degree or order tends to infinity while the inverse approaches an edge;
- (iv) a q -gamma level approaches the minimum while $q \rightarrow 1$.

Problem 14.9 (Canonical inverse double scalings). *Classify the normal forms obtained when two or more derivatives vanish at the same asymptotic rate. Determine when the inverse transition is governed by a fold, cusp, Airy, Pearcey, or Lambert- W coordinate, and derive uniform error bounds across the transition.*

For the last item, if x_q is the minimum point of Γ_q and $Y - \Gamma_q(x_q)$ shrinks together with $1 - q$, the appropriate expansion must combine the q -Stirling correction with a square-root fold. This is a tractable first model.

14.8 Formalization roadmap

The theory separates naturally into layers suitable for mechanized proof.

1. Formal power series: compositional inverse, Bell-polynomial coefficient formulae, and the Puiseux normal form for a finite leading order.
2. Finite algebra: Pochhammer polynomial degrees, roots, derivatives, reciprocity, cyclotomic valuations, and Gaussian positivity.
3. Real analysis: monotonicity intervals, convexity of $\log \Gamma_q$, and uniqueness of real inverse branches.
4. Complex analysis: uniform convergence of logarithmic products, parameter differentiation, and radial root-of-unity asymptotics.
5. Certified numerics: interval enclosures for critical points and branch continuation, producing machine-checkable evidence for the finite-degree conjectures.

The first two layers are largely independent of delicate asymptotics and could be incorporated into the existing semi-formalized corpus early. The inverse Jacobian identities then provide a common API for all parameter inversions.

Chapter 15

Synthesis and practical reference tables

15.1 Which coordinate should be inverted?

Table 15.1: Local inverse coordinates for the principal families.

Forward family	active regime	forward normal form	inverse coordinate
$(a; q)_n$ in a	regular level	$Y - Y_0 \sim c(a - a_0)$	Taylor in $Y - Y_0$
$(a; q)_n$ in a	root collision	$Y \sim c(a - a_0)^m$	$Y^{1/m}$
$(a; q)_n$ in q	$q \rightarrow 1$	$\log(Y/Y_0) \sim K_1 t$	$t = -\log q$
$(a; q)_n$ in q	$q \rightarrow \infty$	$Y \sim Cq^d$	$(Y/C)^{1/d}$
$(a; q)_\infty$ in q	$q = \zeta e^{-t}$	$\log Y \sim -A/t + C$	$t \sim A/(-\log Y + C)$
$(a; q)_\infty$ in $a < 0$	$a \rightarrow -\infty$	$\log Y \sim L^2/(2 \log q)$	$L \sim \sqrt{2 \log q \log Y}$
$(a; q)_\alpha$ in α	$\alpha \rightarrow \infty$	$Y - Y_\infty \sim Cq^\alpha$	$\alpha \sim \log(Y - Y_\infty)/\log q$
$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$ in q	$q \rightarrow 0$	$Y - 1 \sim q$	ordinary reversion
$\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q$ in q	$q \rightarrow 1$	$\log(Y/\binom{n}{k}) \sim c(q - 1)$	logarithmic level
$\Gamma_q(x)$ in x	minimum	$Y - Y_* \sim c(x - x_*)^2$	square root
$B_q(a, s - a)$ in a	center minimum	$\log(Y/Y_*) \sim c(a - s/2)^2$	square root
$e_q((1 - q)z)$ in z	$q \rightarrow 1$	$\log Y \sim z$	Taylor about $\log Y$
$r\phi_s$ in z	first nonzero term z^d	$Y - 1 \sim c_d z^d$	$(Y - 1)^{1/d}$

15.2 A parameter-by-parameter checklist

Given a new q -analogue F , a systematic inverse study should answer the following questions in order.

1. What is the natural domain and which branches of q^x , logarithms, and infinite products define the forward function?
2. Which parameter is active? If several are active, are there enough independent observables for a square inverse system?
3. Where does the corresponding derivative vanish or blow up? Remove all removable singularities before classifying the germ.
4. Which endpoint or singular coordinate regularizes the forward asymptotic: q , $-\log q$, $1/q$, q^α , a logarithmic level, or a fractional power?
5. What are the critical values and how do their cuts label inverse sheets?
6. Is there a real monotonicity or convexity theorem that upgrades a local inverse to a global one?

7. Can a recurrence, logarithmic sum, or scaled representation evaluate the forward function stably on that branch?
8. Which coefficients are proved, which are numerically inferred, and which uniformity assumptions are needed when parameters coalesce?

15.3 Main conclusions

The inverse theory of q -analogues is not a collection of unrelated special-function inverses. It is organized by a small number of normal forms: regular Taylor reversion, algebraic Puiseux ramification, reciprocal inversion at poles, logarithmic endpoint saturation, Laurent–Puiseux inversion at infinity, and $1/\text{Log } Y$ inversion at radial essential singularities. The finite Pochhammer symbol realizes all algebraic forms; the infinite symbol adds logarithmic ends and log-periodicity; Gaussian coefficients add positivity and cyclotomic cancellations; q -gamma and q -beta add convex two-sheeted real geometry; and basic hypergeometric functions assemble these atoms into a broad parameterwise calculus.

The most promising next step is a computer-assisted branch atlas. Exact resultants can locate finite algebraic degeneracies, interval methods can certify real branches, and asymptotic coordinates can seed continuation toward $q = 0, 1, -1, \infty$ and arbitrary roots of unity. Such an atlas would turn the present local theory into a searchable global resource and provide a precise interface with the Fabius–Rvachev and exponent-sequence questions in the surrounding ProveIt corpus.

Appendix A

Universal reversion formulae through fifth order

Let

$$y = A_1x + A_2x^2 + A_3x^3 + A_4x^4 + A_5x^5 + O(x^6), \quad A_1 \neq 0. \quad (\text{A.1})$$

Then

$$\begin{aligned} x = & \frac{y}{A_1} - \frac{A_2}{A_1^3}y^2 + \frac{2A_2^2 - A_1A_3}{A_1^5}y^3 \\ & + \frac{-5A_2^3 + 5A_1A_2A_3 - A_1^2A_4}{A_1^7}y^4 \\ & + \frac{14A_2^4 - 21A_1A_2^2A_3 + 6A_1^2A_2A_4 + 3A_1^2A_3^2 - A_1^3A_5}{A_1^9}y^5 + O(y^6). \end{aligned} \quad (\text{A.2})$$

This single identity produces the origin expansions for finite Pochhammer symbols, Gaussian coefficients, q -polylogarithms, and basic hypergeometric series after the appropriate coefficients are substituted.

For logarithmic forward series

$$u = K_1t + \frac{K_2}{2!}t^2 + \frac{K_3}{3!}t^3 + \frac{K_4}{4!}t^4 + O(t^5), \quad (\text{A.3})$$

substitution into (A.2) gives

$$\begin{aligned} t = & \frac{u}{K_1} - \frac{K_2}{2K_1^3}u^2 + \frac{3K_2^2 - K_1K_3}{6K_1^5}u^3 \\ & + \left(-\frac{K_4}{24K_1^5} + \frac{5K_2K_3}{12K_1^6} - \frac{5K_2^3}{8K_1^7} \right) u^4 + O(u^5). \end{aligned} \quad (\text{A.4})$$

Appendix B

Software inventory and generated artifacts

The reproducibility bundle contains:

`inverse_q_analog_functions.tex`

The complete LaTeX source of this report.

`inverse_q_analog_functions.pdf`

The compiled report.

`inverse_q_analogs_experiments.py`

Commented high-precision numerical checks and figure generation.

`numerical_results.tex`

Generated table included by the main source.

`numerical_results.txt`

Plain-text numerical log.

`finite_pochhammer_branch_atlas.pdf`

Generated branch portrait used in [Figure 13.1](#).

`log_periodic_correction.pdf`

Generated periodic-correction plot used in [Figure 13.2](#).

`qgamma_inverse_branches.pdf`

Generated two-branch plot used in [Figure 13.3](#).

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